

Pleura, lungs, trachea and bronchi

The lungs are the essential organs of respiration and are responsible for the uptake of oxygen into the blood and the removal of carbon dioxide. The functional design of the thorax facilitates this complex process. Acting together, the muscles of respiration and the diaphragm increase the intrathoracic volume, creating a negative pressure within the pleural space that causes lung expansion. The resultant reduction in intra-alveolar pressure prompts the conduction of air through the upper respiratory tract, trachea, bronchi and bronchioles, and finally into the alveoli, where gaseous exchange occurs. The respiratory movements do not clear the lung of its gas with each breath; the presence of a large residual volume of gas minimizes the degree to which each new breath can affect the composition of gas in diffusion exchange with the blood (Lambertsen 1980). Besides efficient oxygen diffusion from alveolar air to blood, the lungs must also provide an *inefficient* carbon dioxide transfer, thus maintaining a stable carbon dioxide blood and tissue pressure, grossly higher (about 40 mmHg) than that in the atmosphere. Maintenance of critical homeostasis for carbon dioxide pressure is of greater importance than precise control of oxygen because so many essential electrical and metabolic functions depend on the hydrogen ion concentration $[H^+]$ of the fluids in which these functions are carried out.

The process of breathing exposes the lung to noxious agents, including gases, dust particles, bacteria and viruses, and to dehydration and freezing. The mucous barrier (including its cellular and immunoglobulin factors), the mucociliary escalator, branching pattern of the airways and the cough reflex are all anatomical defences against these insults. Studies of trends of particle deposition in children and young adults have shown that pulmonary deposition fractions are highest in infants who are at greatest risk from exposure to airborne particulate matter (Asgharian et al 2004). Respiratory function may be compromised either by anatomical defects such as chest wall abnormalities or by paralysis of respiratory muscles. Similarly, ultrastructural abnormalities, such as ciliary dysfunction (as seen in Kartagener's syndrome), lead to recurrent respiratory infections and airway damage.

PLEURA

Each lung is covered by pleura, a serous membrane arranged as a closed invaginated sac. The visceral (pulmonary) pleura adheres closely to the lung surface and follows the interlobar fissures, eventually ensheathing each pulmonary lobe. The visceral pleura continues over the hilum as the parietal pleura, covering the mediastinal organs, most of the diaphragm and the corresponding half of the thoracic wall.

The pleural cavity represents the potential space between the two pleura and incorporates an intervening pellicle of fluid that allows close sliding contact between the two layers during all phases of respiration. This liquid pellicle has a high daily turnover: fluid is produced at 0.01–0.02 ml/kg/h and is continuously absorbed to maintain a pleural fluid level of 0.1–0.2 ml/kg. Fluid movement and absorption are supported by a balance between plasmatic and pleural (hydrostatic and oncotic) pressures and thoracic lymphatic drainage. Pleural effusions occur when the mechanisms that normally resorb the fluid pellicle are destabilized.

The negative pressure developed in the pleural cavity is the result of the opposing outward pull of the chest wall and the inward elastic recoil of the lung. Any change in the elasticity of these structures or accumulation of fluid or air will alter respiratory activity either regionally or globally. Inter-regional disparities in ventilation normally exist as a result of local differences in thoracic expansion and position-related, gravity-dependent gradients in pleural pressure, and are reflected as regional inequalities in gas exchange.

The right and left pleural sacs constitute separate compartments and are contiguous only posterior to the upper half of the sternal body. They come close to each other posterior to the oesophagus at the mid-thoracic level and are widely separate between, where they enclose the

mediastinal space. Cardiac asymmetry dictates that the left pleural cavity is smaller, although it may extend more inferiorly in the mid-axillary line in response to left hemidiaphragmatic disposition. Transudates accumulate earlier within, and have a preference for, the larger right pleural cavity.

The folded portions of the pleura at its reflection sites (retrosternal, interlobar fissures and the azygo-oesophageal recess) are the only aspects of normal pleura that can be visualized radiologically. Demonstration of significant pleural shadowing in any other region usually implies pathological pleural abnormality. At direct inspection of the pleural cavity, either during surgery or thoracoscopy, both parietal and visceral pleura appear translucent; the underlying thoracic muscles and blood vessels are visible beneath the parietal pleura, and the lung and subpleural vascular network are visible beneath the visceral pleura, rendering the latter grey and variegated.

PARIETAL PLEURA

Although it is a continuous structure, different regions of parietal pleura are customarily distinguished by name. Thus the costovertebral pleura lines the internal surface of the thoracic wall and the vertebral bodies, the diaphragmatic pleura lies on the thoracic muscular surface of the diaphragm, the cervical pleura (pleural dome) covers the pulmonary apices, and the mediastinal pleura is applied to the structures between the lungs.

Costovertebral pleura

The costovertebral pleura lines the sternum, ribs, transversus thoracis, intercostal muscles and the sides of the vertebral bodies; normally, it is easily separated from these structures. External to the pleura is a thin layer of loose connective tissue, the endothoracic fascia, analogous to the transversalis fascia of the abdominal wall. Anteriorly, the costal pleura begins posterior to the sternum, where it is continuous with the mediastinal pleura along a line extending from posterior to the sternoclavicular joint inferomedially to the midline posterior to the sternal angle. From here, the right and left costal pleurae descend in contact with each other to the level of the fourth costal cartilages and then diverge. On the right side, the line descends to reach the posterior aspect of the xiphisternal joint, while on the left the line diverges laterally and descends at a distance of 2–2.5 mm from the sternal margin to the sixth costal cartilage, forming the cardiac notch. On each side, the costal pleura sweeps laterally, lining the internal surfaces of the costal cartilages, ribs, transversus thoracis and intercostal muscles. Posteriorly, it passes over the sympathetic trunk and its branches to reach the sides of the vertebral bodies, where it is again continuous with the mediastinal pleura. Superior to the first rib, the costovertebral pleura is continuous with the cervical pleura; inferiorly, it becomes continuous with the diaphragmatic pleura along a line that differs slightly on the two sides. On the right, this line of costodiaphragmatic reflection begins posterior to the xiphoid process, passes posterior to the seventh costal cartilage to reach the eighth rib in the mid-clavicular line and the tenth rib in the mid-axillary line, then ascends slightly to cross the twelfth rib level with the upper border of the twelfth thoracic spine. On the left, the line initially follows the ascending part of the sixth costal cartilage but then follows a course similar to that on the right, although it may be slightly lower.

Diaphragmatic pleura

The diaphragmatic pleura is a thin, tightly adherent layer that covers most of the superior surface of the diaphragm. It is continuous with the costal pleura and medially with the mediastinal pleura along the line of attachment of the pericardium to the diaphragm.

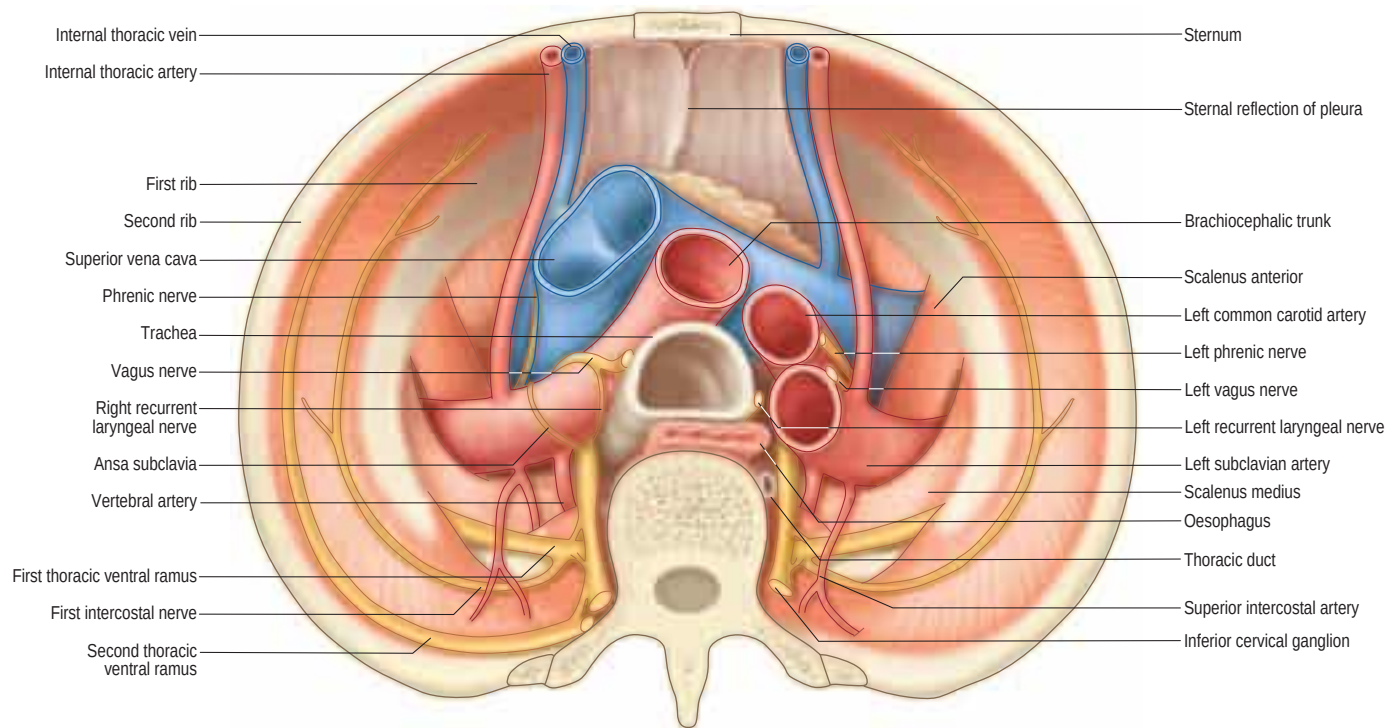


Fig. 54.1 Structures related to the cervical pleura, as seen from below.

Cervical pleura

The cervical pleura is the continuation of the costovertebral pleura over the pulmonary apex, looking as if it is moulded by the latter (see Fig. 51.1). It ascends medially from the internal border of the first rib to the apex of the lung, as high as the inferior edge of the neck of the first rib, and then descends lateral to the trachea to become the mediastinal pleura. As a result of the obliquity of the first rib, the cervical pleura extends 3–4 cm superior to the first costal cartilage, but not superior to the neck of the first rib. The cervical pleura is strengthened by a fascial suprapleural membrane that is attached anteriorly to the internal border of the first rib and posteriorly to the anterior border of the transverse process of the seventh cervical vertebra. It contains a few muscular fibres, which spread from the scaleni. Scalenus minimus extends from the anterior border of the transverse process of the seventh cervical vertebra to the inner border of the first rib behind its subclavian groove, and also spreads into the pleural dome, which it therefore tenses; it has been suggested that the suprapleural membrane is the tendon of scalenus medius. The cervical pleura and the pulmonary apex reach the level of the seventh cervical spine approximately 2.5 cm from the midline. Their surface projection is a curved line from the sternoclavicular joint to the junction of the medial and middle thirds of the clavicle, its summit being 2.5 cm superior to the clavicle (see Fig. 51.4). The subclavian artery ascends laterally in a furrow posterior to the summit of the cervical pleura (Fig. 54.1). The dome is related to the supraclavicular part of the brachial plexus, the stellate ganglion and the vertebral vessels.

Mediastinal pleura

The mediastinal pleura represents the lateral boundary of the mediastinum and forms a continuous coat above the hilum of the lung from the sternum to the vertebral column. On the right, it covers the right brachiocephalic vein, upper part of the superior vena cava, terminal part of the azygos vein, right phrenic and vagus nerves, trachea and oesophagus. On the left, it covers the aortic arch, left phrenic and vagus nerves, left brachiocephalic and superior intercostal veins, left common carotid and subclavian arteries, thoracic duct and oesophagus. At the hilum of the lung, it turns laterally to form a tube that encloses the hilar structures and is continuous with the pulmonary pleura.

VISCERAL PLEURA

The pulmonary pleura is inseparably adherent to the lung over all its surfaces, including those in the fissures, except at the root or hilum of

the lung and along a line descending from this, which marks the attachment of the pulmonary ligament (Figs 54.2–54.3).

INFERIOR PULMONARY LIGAMENTS

The mediastinal pleura extends as a double layer, the pulmonary ligament, inferior to the hilum, from the lateral surface of the oesophagus to the mediastinal surface of the lung, where it is continuous with the visceral pleura (see Figs 54.2B–54.3B). It is continuous superiorly with the pleura around the hilar structures, and inferiorly it ends in a free sickle-shaped contour.

PLEURAL RECESSES

The pleura extends considerably beyond the inferior border of the lung but not as far as the attachment of the diaphragm; therefore, the diaphragm is in contact with the costal cartilages and intercostal muscles below the line of pleural reflection from the thoracic wall to the diaphragm. In quiet inspiration, the inferior margin of the lung does not reach this reflection and the costal and diaphragmatic pleurae are separated merely by a narrow slit, the costodiaphragmatic recess; the lower limit of the lung is normally 5 cm above the lower pleural limit. A similar costomediastinal recess exists posterior to the sternum and the costal cartilages, where the thin anterior margin of the lung falls short of the line of pleural reflection. The extent of this recess, the anterior costomediastinal line of pleural reflection, and the position of the anterior margin of the lung all exhibit individual variation.

The pleural recesses allow the proportionate inspiratory expansion of the lungs, admitting the supplementary volume with minimal friction. The same pellicle of fluid within the recesses separates the parietal costal and diaphragmatic pleura during expiration, or the parietal pleura from the intervening visceral pleura during inspiration. Pleural adhesions and air or fluid accumulation limit this mechanism to a variable degree.

The inferior border of the costodiaphragmatic recess is an important consideration in the surgical posterior approach to the kidney. Usually, the pleura crosses the twelfth rib at the lateral border of erector spinae, so that the medial region of the kidney is above the pleural reflection. The eleventh rib may be mistaken for the twelfth in palpation (sometimes the latter does not project beyond muscle) and an incision prolonged to this level will open the pleura. Whether the lowest palpable rib is the eleventh or twelfth can be ascertained by counting from the second rib (identified at its junction with the sternal angle) or by ultrasound.

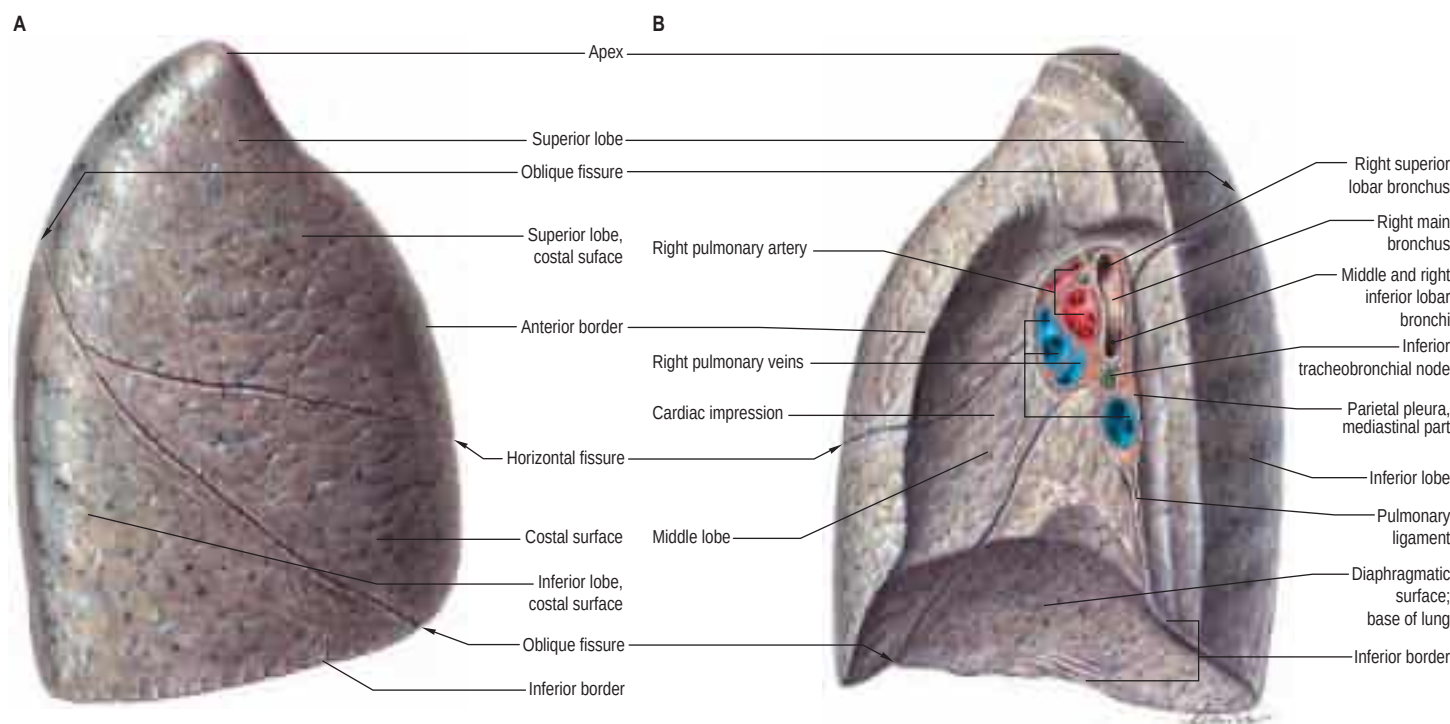


Fig. 54.2 The right lung. **A**, Lateral surface. **B**, Medial surface. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

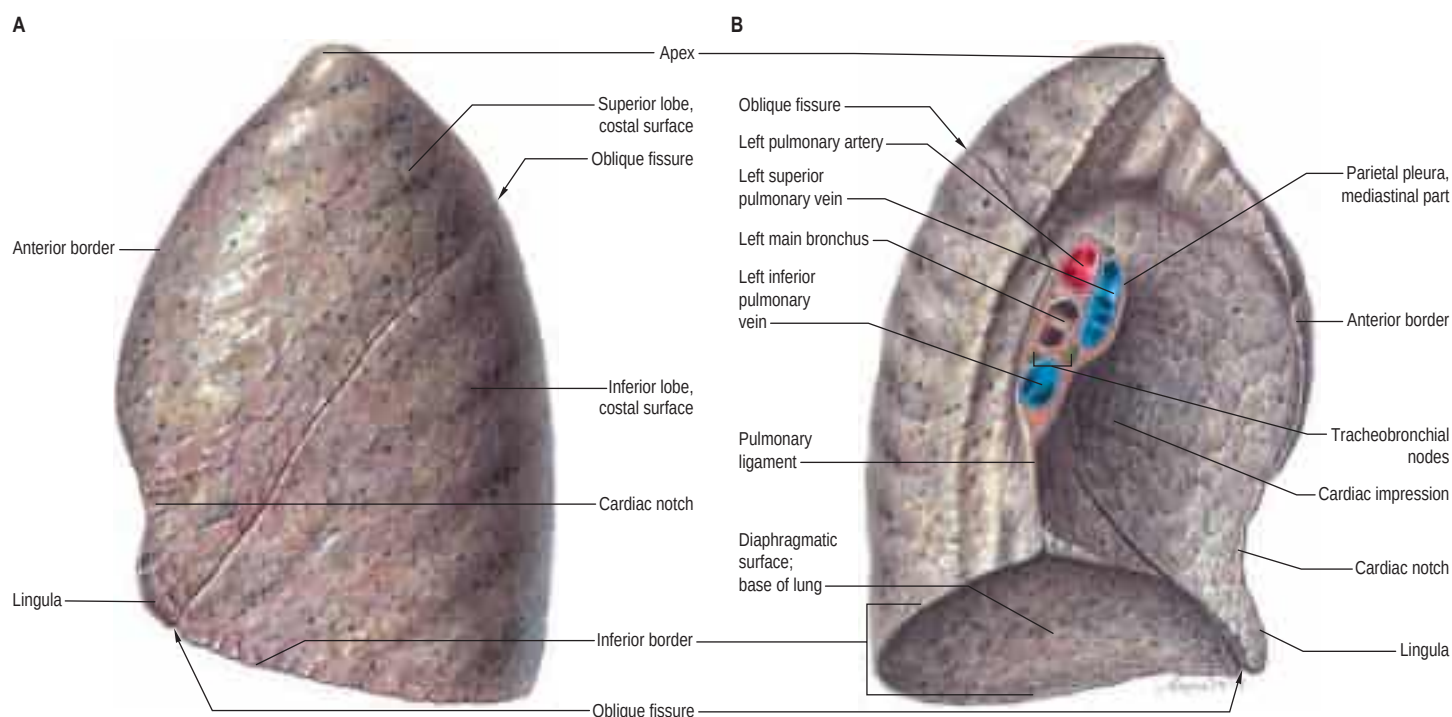


Fig. 54.3 The left lung. **A**, Lateral surface. **B**, Medial surface. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

The parietal and visceral pleurae are developed from somatopleural and splanchnopleural layers of the lateral plate mesoderm, respectively; consequently, the parietal pleura has a somatic arterial supply. The costovertebral pleura is supplied by branches of the intercostal and internal thoracic arteries; the mediastinal pleura by branches from the bronchial, upper diaphragmatic, internal thoracic and mediastinal arteries; the cervical pleura by branches from the subclavian artery; and the diaphragmatic pleura by the superficial part of the diaphragmatic microcirculation. The veins join the thoracic wall veins, eventually draining into the superior vena cava. Lymph from the costovertebral pleura drains into the internal thoracic chain anteriorly

and intercostal chains posteriorly, while that from the diaphragmatic pleura drains into the mediastinal, retrosternal and coeliac axis nodes.

The visceral pleura forms an integral part of the lung and, accordingly, its arterial supply and venous drainage are provided by the bronchial vessels. The bronchial arteries at the hilum form a circle, surrounding the principal bronchus; pleural branches from this anulus supply the visceral pleura facing the mediastinum, the interlobar and apical surfaces, and part of the diaphragmatic surface. The visceral pleura is drained by pulmonary veins, apart from an area around the hilum that drains into bronchial veins. The lymphatic drainage of the visceral pleura is to the deep pulmonary plexus within the interlobar and peribronchial spaces.

INNERVATION

The costal and peripheral diaphragmatic parietal pleurae are innervated by intercostal nerves. The mediastinal and central diaphragmatic parietal pleurae are innervated by the phrenic nerves. Irritation of the former results in pain that is referred to the appropriate part of the thoracic or abdominal wall, whereas irritation of the central diaphragmatic pleura causes pain that is referred to the lower neck and shoulder tip, i.e. to the C3 and 4 dermatomes. Visceral afferents travel along the bronchial vessels with the autonomic fibres.

PNEUMOTHORAX

Any breach of the chest wall and parietal pleura or visceral pleura leads to the accumulation of air within the pleural cavity (pneumothorax). Fluid (hydrothorax), blood (haemothorax) and, rarely, lymph (chylothorax) can also accumulate in this space. Pneumothoraces may occur spontaneously (rupture of emphysematous bullae, or subpleural 'blebs' typically in tall, asthenic males) or following rib or penetrating trauma. Significant air in the pleural space is visible on a chest radiograph because this condition leads to various degrees of pulmonary collapse; there is separation of the parietal and visceral pleurae and an absence of peripheral pulmonary vascular markings in the corresponding area. Occasionally, a valve-like effect occurs, so that air enters the pleural space during inspiration but cannot escape in expiration. This may result in a 'tension pneumothorax', which can be life-threatening and should be suspected whenever there are unilateral decreased breath sounds and hyper-resonance on percussion (due to collapse of the lung), hypotension, jugular venous distension and contralateral tracheal deviation (due to mediastinal shift). A tension pneumothorax requires immediate decompression with intercostal drainage, preceded by, if *in extremis*, a wide-bore needle in the second intercostal space anteriorly in the mid-clavicular line. Fluid collection in the pleural space may be due to congestive cardiac failure, hypoalbuminaemia or inflammatory, infective or neoplastic conditions. An incipient pleural effusion causes obliteration of the costophrenic angle, where a lateral fluid meniscus is often visible from a non-loculated effusion on a frontal chest radiograph. Drainage of the fluid and subsequent analysis are required for diagnostic purposes. Where there is a collection of pus (empyema) or blood, pleural drainage is essential for therapeutic purposes. Ultrasonography is useful in assessing the size and characteristics of an effusion, such as the presence of loculation and debris. It may even demonstrate underlying consolidated lung. Computed tomography (CT) is utilized to assess the underlying lung parenchyma, endobronchium and mediastinal lymph nodes.

MICROSTRUCTURE

The moist, smooth pleural surface consists of a single layer of flat mesothelial (serosal) cells separated from an underlying lamina propria of loose connective tissue by a basal lamina. Ultrastructurally, pleural and peritoneal mesothelial cells are similar in having a highly folded basal plasma membrane, in that their adjacent cell surfaces interdigitate and are joined by desmosomes, and in that their luminal surfaces bear numerous microvilli and some cilia. Cytoplasmic pinocytotic vesicles are common. The connective tissue covers the entire pulmonary surface; it extends from the hilum along the conducting tubes and blood vessels within the substance of the lung and divides the lung into numerous small polyhedral lobules, each of which receives a terminal bronchiole and arteriole, venules, lymphatics and nerves. Lobules vary in size; the superficial ones are the largest, visible as polygonal areas 5–15 mm across.

LUNGS

The lungs are the essential organs of respiration. They are situated on either side of the heart and other mediastinal structures, and occupy most of the thoracic cavity. Each lung is free in its pleural cavity, except at the hilum and pulmonary ligament, where it is attached to the mediastinum. When removed from the thorax, a fresh normal lung is spongy, can float in water, and crepitates when handled because of the air within its alveoli. It is also highly elastic and so retracts on removal from the thorax. Its surface is smooth and shiny, and is divided by fine, dark lines into numerous small polyhedral domains, each crossed by numerous finer lines, indicating the areas of contact between its most peripheral lobules and the pleural surface.

The adult right lung usually weighs 625 g, and the left 565 g, but the range of wet weights is considerable, not least because it reflects the amount of blood or serous fluid contained within the lungs when weighed. In proportion to body stature, the lungs are heavier in men than in women, and boys have bigger lungs than girls (Thurlbeck 1982). Lung volumes in children vary significantly according to race (Sylvester et al 2005), and lung volume, air volume and pulmonary tissue volume all increase linearly with body length (Rao et al 2010). Pulmonary agenesis is a rare congenital abnormality in which pulmonary vessels, bronchi and lung parenchyma are undeveloped; it is associated with congenital heart disease (Nabati et al 2013).

Pulmonary surface features

Each lung has an apex, base, three borders and two surfaces, roughly resembling a halved cone (see Figs 54.2–54.3).

Apex

The apex, the rounded upper extremity, extends above the superior thoracic aperture where it contacts the cervical pleura, and is covered in turn by the suprapleural membrane. As a consequence of the obliquity of the aperture, the apex rises 3–4 cm above the level of the first costal cartilage. The summit of the apex is level posteriorly with the neck of the first rib and lies 2.5 cm above the junction of the mid and medial thirds of the clavicle, therefore within the root of the neck (see Fig. 29.18). It has been claimed that, because the apex does not rise above the neck of the first rib, it is really intrathoracic, and that it is the anterior surface that ascends highest in inspiration. The subclavian artery arches superolaterally over the suprapleural membrane and grooves the anterior surface of the apex near its summit, separating it from scalenus anterior. The cervicothoracic (stellate) sympathetic ganglion, ventral ramus of the first thoracic spinal nerve and superior intercostal artery all lie posterior to the apex. Scalenus medius is lateral and the brachiocephalic trunk, right brachiocephalic vein and trachea are adjacent to the medial surface of the apex of the right lung. The left subclavian artery and left brachiocephalic vein are adjacent to the medial surface of the apex of the left lung.

Base

The basal surface is semilunar and concave, moulded on the superior surface of the diaphragm, which separates each lung from the corresponding lobe of the liver, and the left lung from the gastric fundus and spleen. Since the diaphragm extends higher on the right than on the left, the concavity is deeper on the base of the right lung. Posterolaterally, the base has a sharp margin that projects a little into the costodiaphragmatic recess.

Costal surface

The costal surface of the lung is smooth and convex, and its shape is adapted to that of the thoracic wall, which is vertically deeper posteriorly. It is in contact with the costal pleura; in specimens that have been preserved *in situ*, this surface exhibits grooves that correspond with the overlying ribs, a situation similar to the hyperinflated lungs of asthma patients.

Medial surface

The medial surface has posterior vertebral and anterior mediastinal parts. The vertebral part lies in contact with the sides of the thoracic vertebrae and intervertebral discs, posterior intercostal vessels and splanchnic nerves. The mediastinal area is deeply concave because it is adapted to the heart at the cardiac impression, which is much larger and deeper on the left lung where the heart projects more to the left of the median plane. Posterolaterally to this concavity is the somewhat triangular hilum, where various structures enter or leave the lung, collectively surrounded by a sleeve of pleura that also extends below the hilum and behind the cardiac impression as the pulmonary ligament.

Other impressions on the lung surface

In addition to these pulmonary features, cadaveric lungs that have been preserved *in situ* can show a number of other impressions that indicate their relations with surrounding structures (see Figs 54.2B–54.3B). All these impressions reflect the intimate relationships of the lungs with the adjacent mediastinal structures and offer a good perspective on surgical approaches, either from the anterior or posterior aspect of the corresponding pleural cavity.

On the right lung, the cardiac impression is related to the anterior surface of the right auricle, the anterolateral surface of the right atrium and partially to the anterior surface of the right ventricle. The

impression ascends anterior to the hilum as a wide groove for the superior vena cava and the terminal portion of the right brachiocephalic vein. Posteriorly, this groove is joined by a deep sulcus that arches forwards above the hilum and is occupied by the azygos vein. The right side of the oesophagus makes a shallow vertical groove behind the hilum and the pulmonary ligament. Towards the diaphragm it inclines left and leaves the right lung, and therefore does not reach the lower limit of this surface. Posteroinferiorly, the cardiac impression is confluent with a short, wide groove adapted to the inferior vena cava. Between the apex and the azygos groove, the trachea and right vagus are close to the lung but do not mark it.

On the left lung, the cardiac impression is related to the anterolateral surface of the left ventricle and auricle (see Fig. 54.3B). The anterior infundibular surface and adjoining part of the right ventricle are related to the lung as it ascends in front of the hilum to accommodate the pulmonary trunk. A large groove arches over the hilum, and descends behind it and the pulmonary ligament, corresponding to the aortic arch and descending aorta. From its summit, a narrower groove ascends to the apex for the left subclavian artery. Posterior to this, and superior to the aortic groove, the lung is in contact with the thoracic duct and oesophagus. Anterior to the subclavian groove there is a faint linear depression for the left brachiocephalic vein. Inferiorly, the oesophagus may mould the surface anterior to the lower end of the pulmonary ligament.

Pulmonary borders

The inferior border is thin and sharp where it separates the base from the costal surface and extends into the costodiaphragmatic recess, and more rounded medially where it divides the base from the mediastinal surface. It corresponds, in quiet respiration, to a line drawn from the most inferior point of the anterior border that passes to the sixth rib at about the mid-clavicular line, then to the eighth rib in the mid-axillary line (usually 10 cm above the costal margin), and then continues posteromedially and slightly superiorly to a point 2 cm lateral to the tenth thoracic spine (see Fig. 51.5). The diaphragm rises more superiorly on the right to accommodate the liver, and so the right lung is vertically shorter (by approximately 2.5 cm) than the left. However, cardiac asymmetry means that the right lung is broader, and has a greater capacity and weight than the left. The posterior border separates the costal surface from the mediastinal surface, and corresponds to the heads of the ribs. It has no recognizable markings and is really a rounded junction of costal and vertebral (medial) surfaces. The thin, sharp, anterior border overlaps the pericardium. On the right, it corresponds closely to the costomediastinal line of pleural reflection, and is almost vertical. On the left it approaches the same line above the fourth costal cartilage; below this point, it shows a variable cardiac notch, the edge of which passes laterally for 3.5 cm before curving inferomedially to the sixth costal cartilage 4 cm from the midline. It therefore does not reach the line of pleural reflection here (see Fig. 51.4) and so the pericardium is covered only by a double layer of pleura (area of superficial cardiac dullness). Surgical experience suggests that the line of pleural reflection, the anterior pulmonary margin and the costomediastinal pleural recess are all variable.

PULMONARY FISSURES AND LOBES

Right lung

The right lung is divided into superior, middle and inferior lobes by its oblique (major) and horizontal (minor) fissures (see Fig. 54.2). The superior, lengthier oblique fissure separates the inferior from the superior and middle lobes, and corresponds closely to the left oblique fissure, although it is less vertical, crossing the inferior border of the lung approximately 7.5 cm behind its anterior end. On the posterior border, it is either level with the spine of the fourth thoracic vertebra or slightly lower. It descends across the fifth intercostal space and follows the sixth rib to the sixth costochondral junction. The shorter horizontal fissure separates the superior and middle lobes. It passes from the oblique fissure, near the mid-axillary line, horizontally forwards to the anterior border of the lung, level with the sternal end of the fourth costal cartilage, then passes posteriorly to the hilum on the mediastinal surface. The horizontal fissure is usually visible on a frontal chest radiograph. The oblique fissure is usually visible on a lateral radiograph and on a high-resolution CT scan as a curvilinear band from the lateral aspect to the hilum (Fig. 54.4). The smaller middle lobe is wedged between the superior and inferior lobes, and includes some

of the costal surface, the inferior part of the anterior border and the anterior part of the base of the lung. The azygos vein may sometimes course in a more lateral position, within a four-layered pleural septum within the superior lobe, creating an 'azygos lobe'. This is not a true lobe, but a separated portion of the superior lobe that lacks its own bronchi, arteries or veins. The condition represents an incidental finding on chest radiographs or CT scans, with no associated morbidity; it may be confused with a pulmonary nodule, and loculated effusions in the fissure may appear. Accessory fissures may separate either the medial basal segment (Twining's line) or the superior segment from the remainder of the inferior lobe. Incomplete separation of the lobes and identification of the completeness of the fissures are important prior to lobectomy because individuals with incomplete fissures are more prone to develop postoperative air leaks and may require further procedures, such as stapling and pericardial sleeves.

Left lung

The left lung is divided into superior and inferior lobes by its oblique fissure, which extends from the costal to the medial pulmonary surfaces, superior and inferior to the hilum (see Fig. 54.3). Superficially, this fissure begins on the medial surface at the posteriosuperior part of the hilum, ascends obliquely and posteriorly to cross the posterior border of the lung 6 cm below the apex, then descends anteriorly across the costal surface, to reach the inferior border almost at its anterior end. It finally ascends on the medial surface to the inferior part of the hilum. At the posterior border of the lung, the fissure usually lies opposite a surface point 2 cm lateral to the midline between the spines of the third and fourth thoracic vertebrae, but it may be superior or inferior to this level. Traced around the chest, the fissure reaches the fifth intercostal space (at or near the mid-axillary line) and follows this to intersect the inferior border of the lung, close or just inferior to the sixth costochondral junction (7.5 cm from the midline). The left oblique fissure is usually more vertical than the right, and is indicated approximately by the medial border of the scapula with the fully abducted arm. A left horizontal fissure is a normal, occasional variant.

The superior lobe lies anterosuperior to the oblique fissure and includes the apex, anterior border, much of the costal and most of the medial surfaces of the lung. A small process, the lingula, is usually present at the lower end of the cardiac notch. The larger inferior lobe lies behind and below the fissure, and contributes almost the whole of the base, much of the costal surface and most of the posterior border of the left lung.

Lung lobation, and the morphology of the tracheobronchial tree, can predict the atrial situs.

In situs solitus, the right principal bronchus is short and eparterial (its branch for the right upper lobe lies over the second branch of the right pulmonary artery) and the left principal bronchus is longer and hyparterial (it courses underneath the left pulmonary artery). A bilobed left lung and a trilobed right lung are typical in situs solitus. In situs inversus, there is a mirror imaging of the disposition seen in situs solitus. In situs ambiguus, the bronchi and lungs can display either a bilateral right morphology with bilateral trilobed lungs and bilateral eparterial bronchi (heterotaxy syndrome: incomplete or inappropriate lateralization of the thoracic and abdominal viscera, and asplenia) or a bilateral left morphology with bilateral bilobed lungs and bilateral hyparterial bronchi (heterotaxy syndrome and polysplenia) (Van Praagh et al 1992).

Bronchopulmonary segments

Each of the principal bronchi divides into lobar bronchi. Primary branches of the right and left lobar bronchi are termed segmental bronchi because each ramifies in a structurally separate, functionally independent, unit of lung tissue called a bronchopulmonary segment (Figs 54.5–54.7).

The main segments are named and numbered as shown in Table 54.1.

Each segment is surrounded by connective tissue that is continuous with the visceral pleura, and is a separate respiratory unit. The vascular and lymphatic arrangements of the segments are described on page 959.

PULMONARY HILA

The pulmonary root connects the medial surface of the lung to the heart and trachea, and is formed by a group of structures that enter or leave

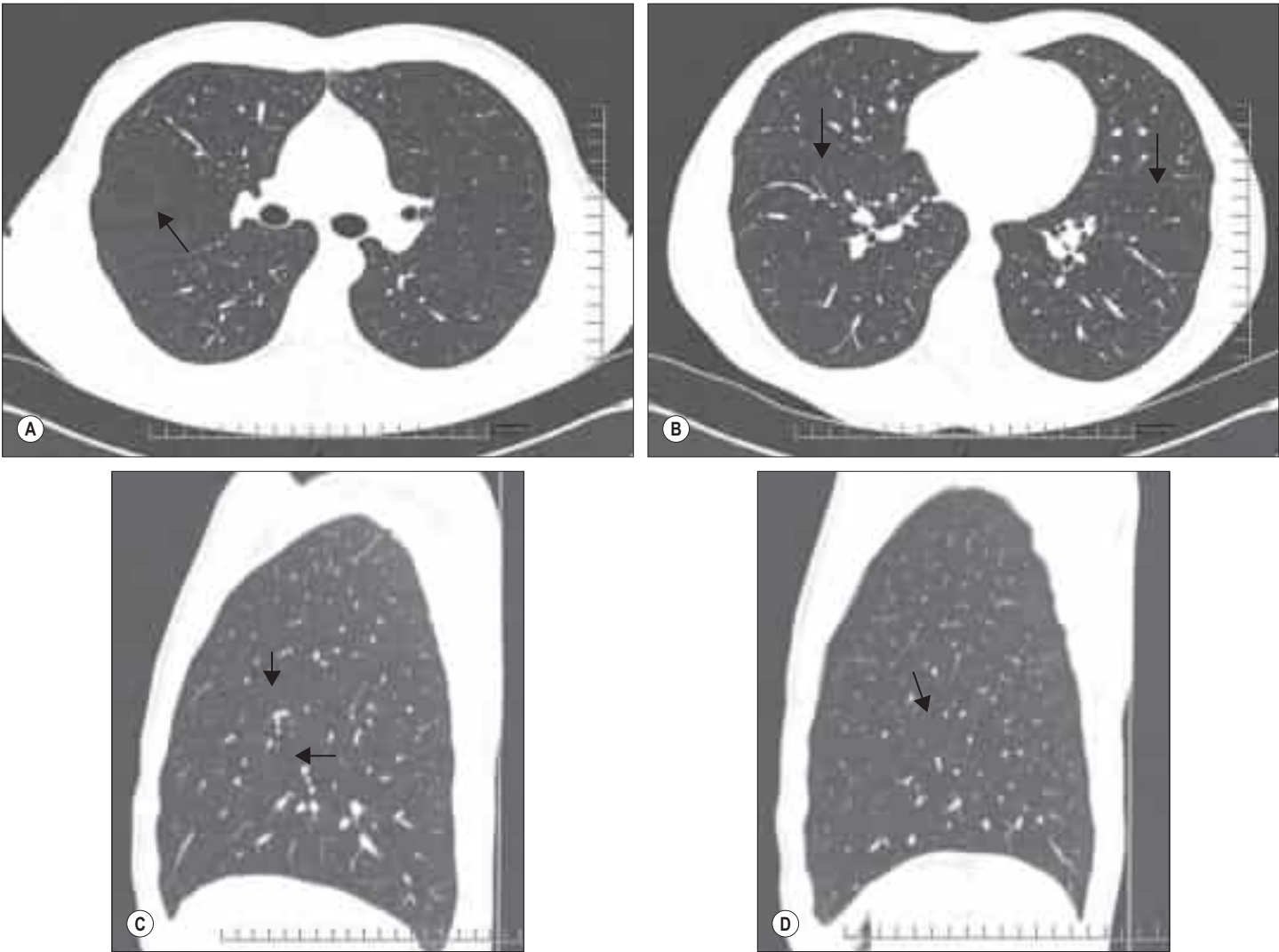


Fig. 54.4 **A**, An axial section through the lungs below the carina, demonstrating the left and right oblique fissures. The circular avascular area anterior to the right oblique fissure and surrounded by a halo of white (arrow) is due to the horizontal fissure. In this individual, the fissure is slightly domed and passes through the plane of the section. **B**, An axial section through the lungs at the mid left atrial level, demonstrating the left and right oblique fissures (arrows) as they pass more anteriorly and separate the lower lobes from the lingula and middle lobes. **C**, A sagittal section through the right lung, demonstrating the oblique (down-pointing arrow) and horizontal fissures (horizontal arrow). **D**, A sagittal section through the left lung, demonstrating the oblique fissure (arrow).

Table 54.1 Bronchopulmonary segments

	Right lung	Left lung
Superior lobe	I, apical; II, posterior; III, anterior	I, apical; II, posterior; III, anterior; IV, superior lingular; V, inferior lingular
Middle lobe	IV, lateral; V, medial	—
Inferior lobe	VI, superior (apical); VII, medial basal; VIII, anterior basal; IX, lateral basal; X, posterior basal	VI, superior (apical); VII, medial basal; VIII, anterior basal; IX, lateral basal; X, posterior basal

the hilum (see Figs 54.2B–54.3B, 54.6C,D). These are the principal bronchus, pulmonary artery, two pulmonary veins, bronchial vessels, a pulmonary autonomic plexus, lymph vessels, bronchopulmonary lymph nodes and loose connective tissue, all enveloped by a pleural sleeve. The pulmonary roots, or pedicles, lie opposite the bodies of the fifth to seventh thoracic vertebrae. The phrenic nerve, pericardiophrenic vessels and anterior pulmonary plexus are common anterior relations of both hila. The vagus nerve and posterior pulmonary plexus are common posterior relations and the pulmonary ligament is a common inferior relation. The major structures in both roots are similarly arranged: the superior of the two pulmonary veins is anterior, the pulmonary artery and principal bronchus are more posterior, and the bronchial vessels are most posterior. The arrangement of bronchopulmonary segments and the pulmonary hila permit the resection of abscesses and localized primary lung malignancy.

Right hilum

The right root is situated behind the superior vena cava and right atrium, inferior to the terminal part of the azygos vein. The usual sequence of hilar structures, from superior to inferior, is superior lobar (eparterial) bronchus, pulmonary artery, principal bronchus and inferior pulmonary vein (see Fig. 54.2B).

Left hilum

The left root lies inferior to the aortic arch and anterior to the descending thoracic aorta. The usual vertical sequence of structures at the left hilum is pulmonary artery, principal (hyparterial) bronchus and inferior pulmonary vein (see Fig. 54.3B). The pulmonary artery is longer on the left side, and each of its branches from the hilum to the oblique fissure must be identified in pulmonary resections.

SECONDARY PULMONARY LOBULES

Each segmental bronchus supplies a bronchopulmonary segment. Progressive subdivisions of the bronchi occur within each segment, and the bronchi become increasingly narrow. All intrapulmonary bronchi are kept patent by cartilaginous plates that decline in size and number, and finally disappear when bronchi become bronchioles (less than 1 mm in diameter). The secondary lobule is the smallest macroscopic subsection of the peripheral lung, bounded by connective tissue septa of

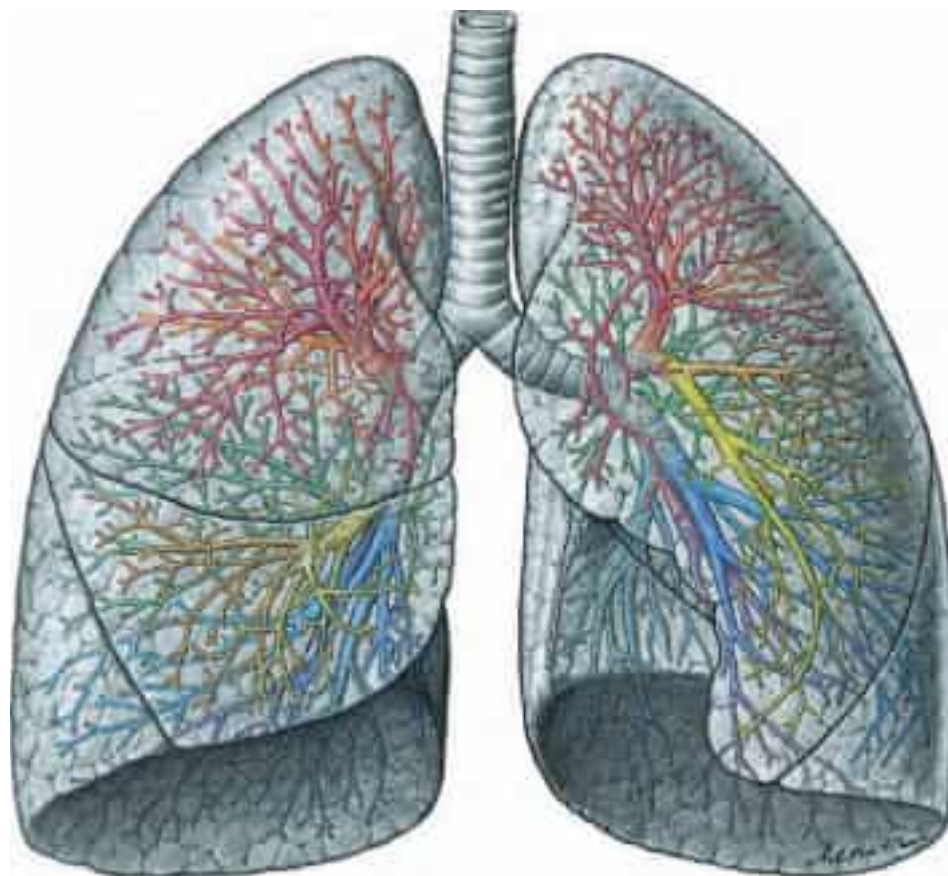


Fig. 54.5 Lungs and bronchi: anterior aspect. The lobar and segmental bronchi are projected on to the lung in different colours (compare with Fig. 54.6). (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

uneven size and shape. It consists of approximately six terminal bronchioles, the latter being the most peripheral bronchioles not to have alveoli in their walls. Distal to each terminal bronchiole is an acinus, the functional unit of the lung for gas exchange, which consists of 3–4 orders of respiratory bronchioles, each leading to 3–8 orders of alveolar ducts. The walls of these ducts consist of alveolar sacs or the mouths of alveoli. The primary lobule is the lung distal to the respiratory bronchiole.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

The lungs have two functionally distinct circulatory pathways. The pulmonary vessels convey systemic deoxygenated blood to the alveolar walls and drain oxygenated blood back to the left atrium for systemic distribution by the left ventricle. The much smaller bronchial vessels, which are derived from the systemic circulation, provide oxygenated blood to lung tissues that do not have close access to atmospheric oxygen, i.e. those of the bronchi and larger bronchioles.

The pulmonary trunk bifurcates into right and left pulmonary arteries that approach the hila of the lungs. Each artery divides into branches that accompany the segmental and subsegmental bronchi, mostly in a posterolateral position. The pulmonary capillaries form single-layered and exceedingly thin-walled plexuses immediately outside the epithelium in the walls and septa of the alveoli and alveolar sacs. Pulmonary veins, two from each lung, drain the pulmonary capillaries; their rootlets coalesce into larger and more freely communicating veins that traverse the lung, for the most part independently of the pulmonary arteries and bronchi. At the hilum, the pulmonary vessels accompany the main bronchial divisions; the veins are anteromedial and the arteries posterolateral to the bronchi. This is not the case in the bronchopulmonary segments, where a segmental bronchus, its branches and associated arteries occupy a central position but the many tributaries of the pulmonary veins run between segments, serving adjacent segments (which therefore drain into more than one vein). Some veins also lie beneath the visceral pleura, including the pleura in the interlobar fissures. It follows from this that a bronchopulmonary segment is not a complete vascular unit with an individual bronchus, artery and vein. During resection of segments, it is obvious that the planes between them are not avascular but are crossed by pulmonary veins and sometimes by branches of arteries. This pattern of bronchi, arteries and veins exhibits considerable variation; veins are the most variable, and arteries are more variable than bronchi.

Pulmonary arteries

The right pulmonary artery divides into two large branches as it emerges behind the superior vena cava (Fig. 54.8). This intimate relationship facilitates the cavopulmonary anastomosis (the Glenn shunt) conceived to bypass the defective right heart chambers in various congenital cardiac disorders. A lymph node usually lies adjacent to the bifurcation. The smaller branch to the superior lobe usually divides again, supplying the majority of that lobe. The inferior branch descends anterior to the bronchus intermedius (see below) and posterior to the superior pulmonary vein. It provides a small recurrent branch to the superior lobe, and then, at the point where the horizontal fissure meets the oblique fissure, it gives off the branch to the middle lobe anteriorly, and a branch to the superior segment of the inferior lobe posteriorly. It then continues a short distance before dividing to supply the rest of the inferior lobe segments.

The left pulmonary artery emerges from within the concavity of the aortic arch and descends anterior to the descending aorta to enter the oblique fissure (see Fig. 54.8). The ligamentum arteriosum skirts between the superior aspect of the left pulmonary artery and the proximal descending aorta. The left recurrent laryngeal nerve winds around the aorta, to the left of the ligament; the aorta is relatively fixed by the ligament, and so may be ruptured in cases of major trauma associated with rapid deceleration. The branches of the left pulmonary artery are extremely variable. The first and largest branch is usually given off to the anterior segment of the left superior lobe. Before reaching the oblique fissure, the artery gives off a variable number of other branches to the superior lobe, and, as it enters the fissure, it usually supplies a large branch to the superior segment of the inferior lobe. Lingular branches arise within the fissure, and the rest of the lower lobe is supplied by many varied branching patterns. It was a surgical aphorism of the late Lord Brock that when performing a left upper lobectomy, 'there was always one more branch of the pulmonary artery than you thought!'

Pulmonary sequestration

When a portion of lung exists without the appropriate bronchovascular connections, it is usually supplied by the systemic vasculature (thoraco-abdominal aorta or branches) and drains into either atrium, superior vena cava or azygos veins (Clements 1999) (Figs 54.9–54.10). Extralobar pulmonary sequestration segments are covered by visceral pleura and usually found below the left inferior lobe, whereas intralobar abnormalities are usually embedded in normal lung; classically, the posterior basal segment of the left inferior lobe. Extralobar pulmonary



Unilateral absence of a pulmonary artery is a rare congenital abnormality characterized by normal lung volume and anatomy at birth. Revascularization before the age of 6 months avoids the development of lung hypoplasia ([Alison et al 2011](#)). The left pulmonary artery sling is a congenital abnormality characterized by the left pulmonary artery arising from the right pulmonary artery, coursing over the right principal bronchus and heading posteriorly between the trachea and oesophagus. This abnormality is associated with significant tracheobronchial stenosis ([Zhong et al 2010](#)).

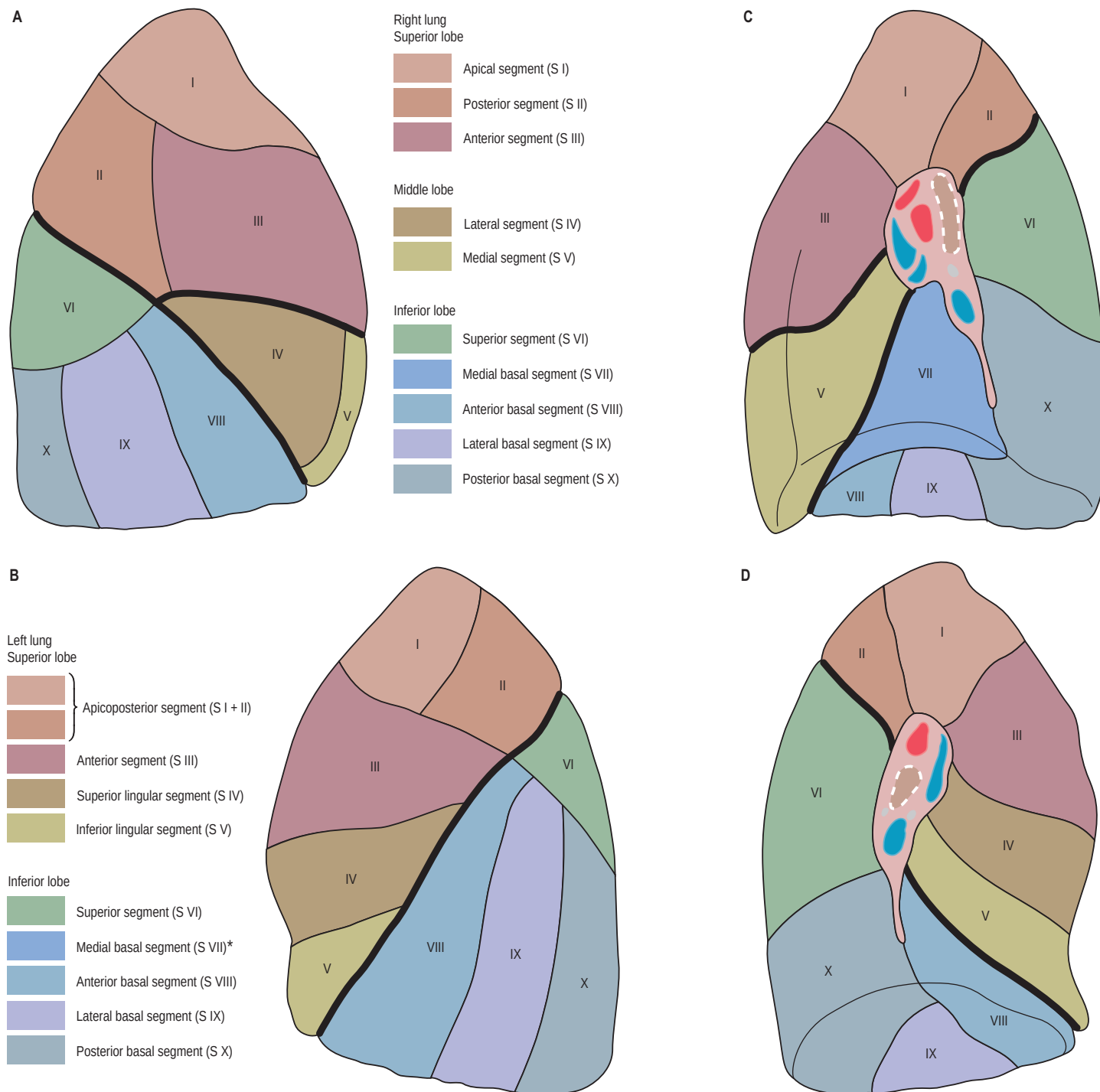


Fig. 54.6 Bronchopulmonary segments of the right and left lungs, coloured to indicate the different territories. **A**, Right lung, lateral aspect. **B**, Left lung, lateral aspect. **C**, Right lung, medial aspect. **D**, Left lung, medial aspect. *Usually, this segment is not a separate unit but is fused with the anterior basal segment. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

sequestration *per se* is asymptomatic but is often associated with other congenital abnormalities: pulmonary hypoplasia, congenital diaphragmatic hernia, congenital cystic adenomatoid malformation and congenital heart disorders (Stocker et al 1978, Kravitz 1994). A fistula with the oesophagus may also develop (Ch. 52). Consideration of differential diagnosis with mediastinal or pulmonary tumours is mandatory.

Pulmonary embolism and arteriography



Available with the *Gray's Anatomy* e-book

Pulmonary veins

There are usually four pulmonary veins, two from each lung. They originate from capillary networks in the alveolar walls and return oxygenated blood to the left atrium. All the main tributaries of the pulmonary veins receive smaller tributaries, both intra- and intersegmental; by serial

junctions, tributary veins finally form a single lobar trunk, i.e. three in the right lung, two in the left. The right middle and superior lobar veins usually unite and so two veins, superior and inferior, leave each lung.

At the pulmonary hilum, the usual distribution of veins is as follows. The superior pulmonary vein is antero-inferior to the pulmonary artery, and the inferior pulmonary vein is the most inferior hilar structure and also slightly posterior. On the right, the union of apical, anterior and posterior veins (draining the upper lobe) with a middle lobar vein formed by lateral and medial tributaries constitutes the right superior pulmonary vein. The right inferior pulmonary vein is formed by the hilar union of superior (apical) and common basal veins from the lower lobe. The union of superior and inferior basal tributaries forms the common basal vein. Occasionally, the three right lobar veins remain separate. The right superior pulmonary vein passes posterior to the superior vena cava, the inferior behind the right atrium.

The superior left pulmonary vein is formed by the union of apico-posterior, anterior and lingular veins. The left inferior pulmonary vein is formed from the union of the superior (apical) and common basal

A thrombus that has developed in the deep veins (usually leg or pelvic) may embolize and travel in the right side of the circulation through the right atrium and ventricle, lodging in the pulmonary vasculature. The clinical abnormalities observed depend on the size of the embolus and on the number and frequency of embolic episodes. Large emboli may lodge in the main pulmonary artery branches and cause right ventricular dysfunction and hypoxia, representing a medical emergency. Smaller emboli may lodge in segmental pulmonary arteries and cause pleuritic chest pain, shortness of breath and haemoptysis but the clinical picture is rarely pathognomonic. Pulmonary emboli cause a ventilation/perfusion mismatch that can have serious physiological implications leading to a significant reduction in the oxygenation of

blood. Ventilation/perfusion scans with radiolabelled xenon and technetium usually demonstrate segmental abnormalities in perfusion with normal ventilation in the corresponding regions. Pulmonary emboli may also be evaluated by contrast-enhanced spiral CT or pulmonary angiograms and appear as filling defects. Repetitive embolization may eventually lead to a dramatic reduction of the pulmonary vascular bed and chronic cor pulmonale. Thrombolysis, anticoagulation or inferior vena caval filters prevent progression or recurrence of embolism. A clot may also traverse a patent foramen ovale and lodge in the arterial system (paradoxical embolism), most severely in the cerebral circulation, causing a stroke (Kent and Thaler 2011).



Fig. 54.9 A lung window axial view showing an intralobar sequestration in the left lower lobe of the lung in a 15-month-old girl. (Courtesy of Mr Hany Gabra & Mr Jonathan Goring, Great North Children's Hospital, Newcastle upon Tyne, UK.)

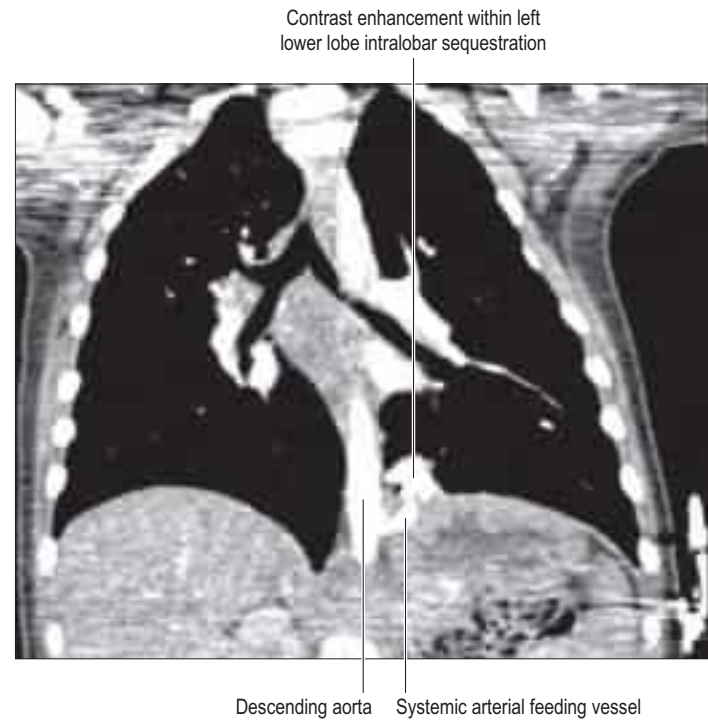


Fig. 54.10 The systemic arterial blood supply from the abdominal aorta to an intralobar sequestration in the left lower lobe of the lung in a 15-month-old girl. (Courtesy of Mr Hany Gabra & Mr Jonathan Goring, Great North Children's Hospital, Newcastle upon Tyne, UK.)

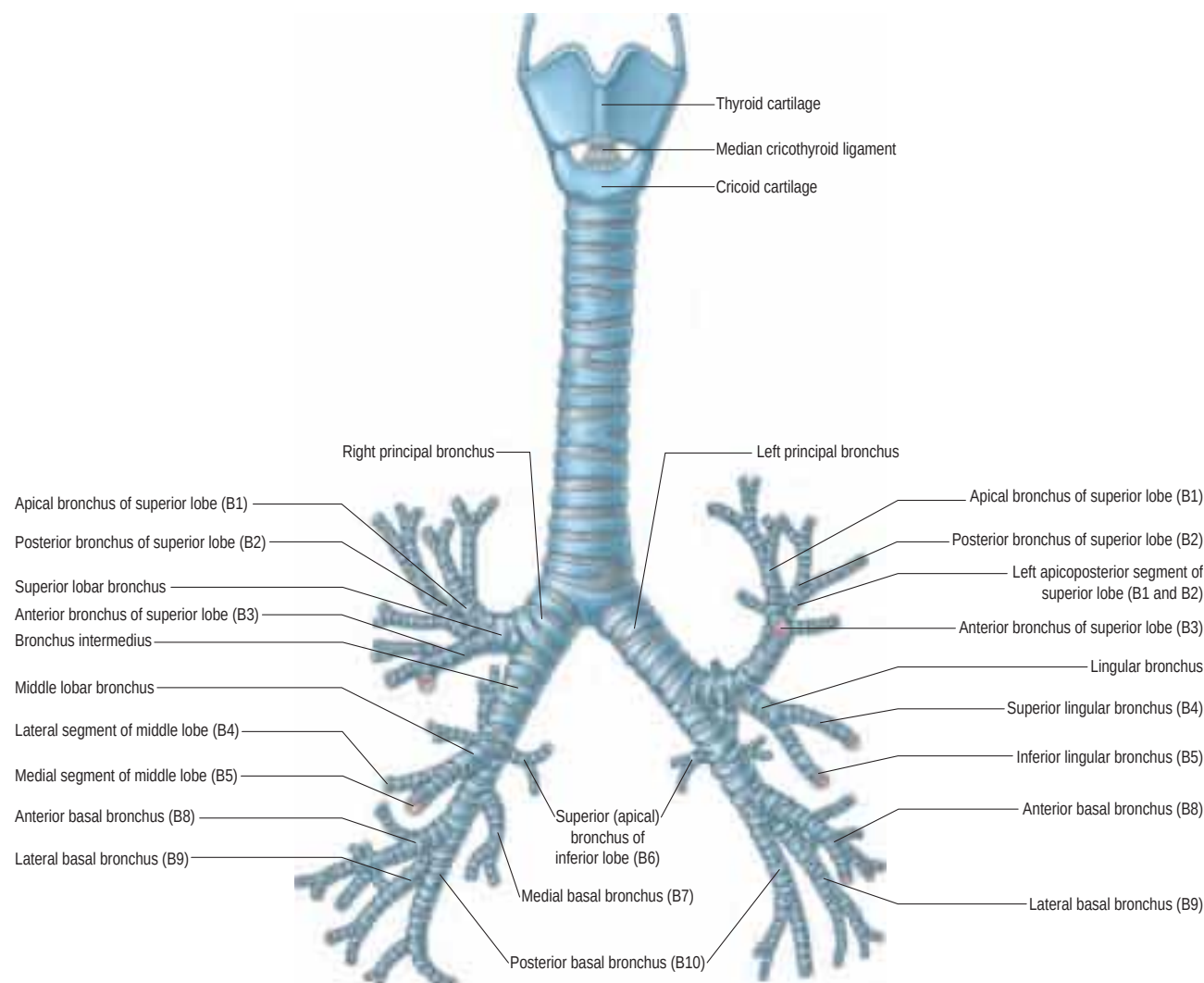


Fig. 54.7 The cartilages of the larynx, trachea and bronchi, anterior aspect. The bronchopulmonary segments are shown in brackets. (Redrawn from an original figure drawn from a metal cast made by the late Lord Russell Brock, GKT School of Medicine, London.)

veins, the latter formed by the union of a superior and an inferior basal vein. Both left superior and inferior pulmonary veins pass anterior to the descending thoracic aorta. Sometimes, the two left pulmonary veins form a single trunk, or they may be augmented by an accessory lobar vein from each lobe, which unite to form a third left pulmonary vein.

The right and left pulmonary veins perforate the fibrous pericardium and open separately in the posterosuperior aspect of the left atrium (see Figs 57.3, 57.4B). Their terminations are separated medially by the oblique pericardial sinus, and laterally by smaller and variable pulmonary venous pericardial recesses that are directed superomedially. The pulmonary veins are devoid of valves.

The variable incorporation of the primitive pulmonary vein into the left atrium means that the number of main pulmonary veins may differ. The left atrium is the heart chamber most posterior and median in position, and so the length of the left and right pulmonary veins is equal, unlike the arteries and bronchi, where there are conspicuous left-right differences in length. The terminal parts of the pulmonary veins are surrounded by atrial myocardium; these areas represent potential accessory re-entrant circuits responsible for the initiation or maintenance of supraventricular tachycardias or atrial fibrillation and may be percutaneously ablated. Pulmonary vein atresia is a rare congenital abnormality (Kozak et al 2011).

Lymphatic drainage

The pulmonary lymphatics originate in superficial and deep plexuses. The superficial plexus lies deep to the visceral pleura; its efferents turn around the margins of the lung and its fissures, eventually reaching the bronchopulmonary nodes. The alveoli have no lymph vessels. The deep lymphatics originate at bronchiolar level and follow the broncho-vascular divisions to the hilum, ending in the same nodes. No free anastomoses exist between the two systems, except at the hilum and peripherally, where small but dilatable channels connect superficial and

deep lymphatic vessels; these connections direct lymph from the deep to the superficial channels when outflow from deep vessels is obstructed by pulmonary disease. There is a tendency for vessels from the superior lobes to pass to the superior tracheobronchial nodes, and those from the inferior lobes to pass to the inferior tracheobronchial group. However, lymphatic vessels of adjoining lobes connect deep in the fissures and so these connections are not exclusive. At the level of pulmonary lobation, the arrangement of lymphatic vessels follows the central artery of a lobule and its peripheral veins. Lymphoid aggregations, non-follicular in appearance, occur in peribronchial sites and in 'placoid' formations deep to the pulmonary pleura.

INNERVATION

The autonomic nervous system controls many aspects of airway function, including regulation of airway smooth muscle tone, mucus secretion from submucosal glands and surface epithelial goblet cells, vascular permeability and blood flow (Belvisi 2002).

Pulmonary plexuses

The pulmonary plexuses lie anterior and posterior to the other hilar structures of the lungs (see [Figs 56.7, 57.61](#)). The anterior plexus is small and is formed by rami from vagal and sympathetic cervical cardiac nerves via connections with the superficial cardiac plexus. The posterior pulmonary plexus is formed by the rami of vagal and sympathetic cardiac branches from the second to fifth or sixth thoracic sympathetic ganglia. The left plexus also receives branches from the left recurrent laryngeal nerve. Further details of the pulmonary plexuses are given in the description of the cardiac plexuses on [page 1021](#). The sympathetic nervous system (noradrenaline (norepinephrine)) acting on β -receptors produces bronchodilation; the parasympathetic nervous system (acetylcholine) acting on muscarinic M3 receptors maintains

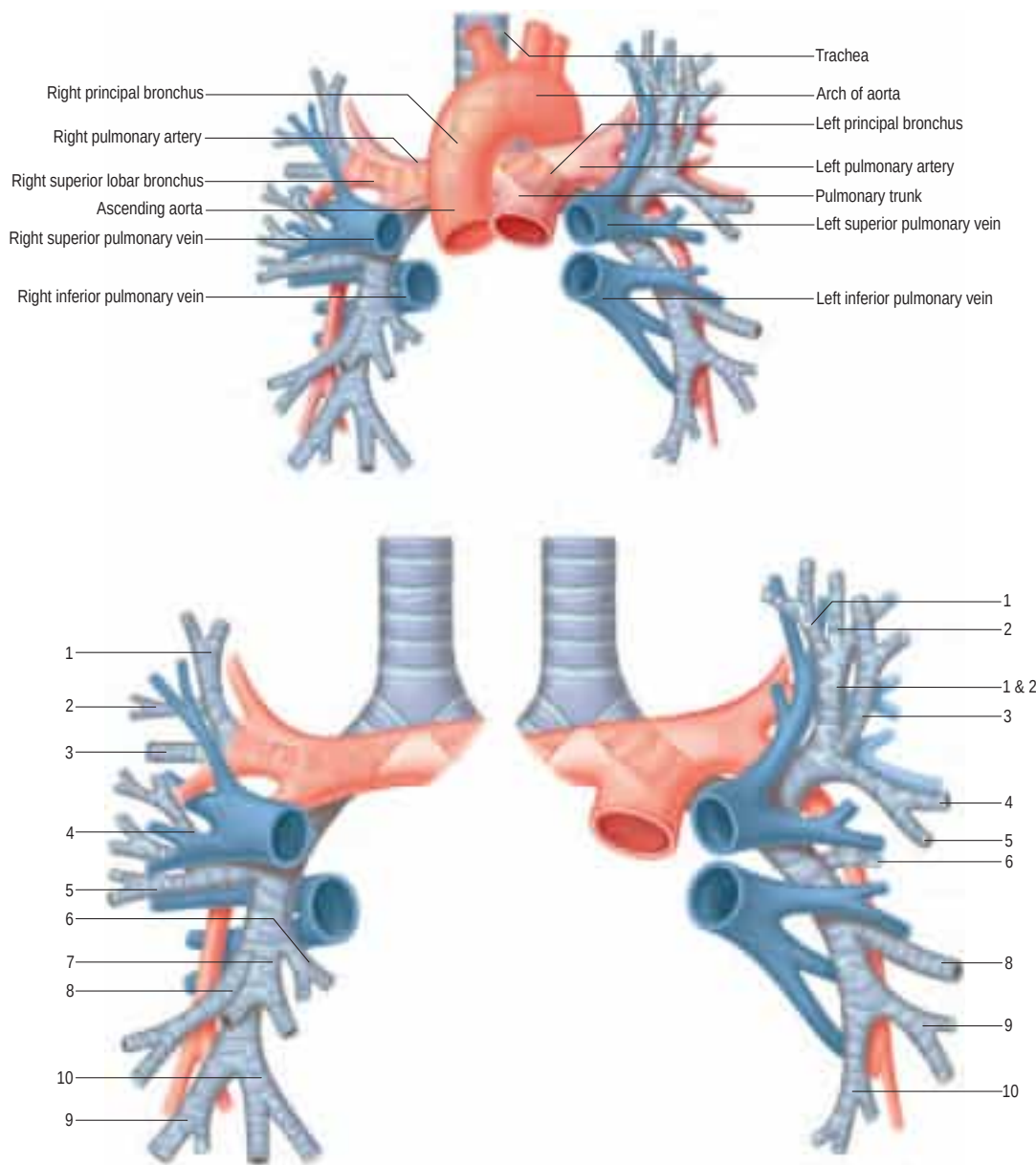


Fig. 54.8 The relationship between the central airways and the pulmonary vessels. The numbers in the lower figure denote the numbers of the bronchopulmonary segments.

the resting tone of the bronchiolar smooth muscle. Prostaglandins have various effects: PGI_2 produces bronchodilation (acting on IP_1 receptors) or bronchoconstriction (acting on EP_1 receptors); $\text{PGF}_{2\alpha}$ produces bronchoconstriction.

MICROSTRUCTURE OF RESPIRATORY SURFACES

Thin-walled respiratory surfaces (alveoli) are distributed as isolated patches within the walls of respiratory bronchioles and as tube-like alveolar ducts and balloon-like alveolar sacs that contain groups of adjacent alveoli.

Alveolar area

Both human lungs contain 300–500 million alveoli, giving a total surface area of 70–100 m^2 . These values vary considerably between normal young individuals; the differences become even more marked with age and as a consequence of degenerative changes. The number of alveoli increases from approximately 10 million at birth to the adult value, mainly in the first 8 years; thereafter, the increase in lung volume is achieved through augmentation of alveolar volume (Narayanan et al 2013). Their inflated diameter varies with lung position, and is greater in the superior regions than in the inferior because of the increased gravitational pressure at the lung base. Although the lung enlarges in all directions during inspiration, most of the change in volume appears to be due to the lengthening and expansion of the alveolar ducts and increase in the size of the openings into the air sacs; the inspiratory change in the volume of the gas-conducting system is insignificant. The combined alveolar and capillary surface area for gas exchange is

60–80 m^2 surface. The capillary blood transit time at rest is 0.75 s, halving with exercise.

Alveolar structure

The alveoli are thin-walled pouches that provide the respiratory surface for gaseous exchange (Figs 54.11–54.13). Their walls contain two types of epithelial cell (pneumocytes) and cover a delicate connective tissue within which a network of capillaries ramifies. Since the walls are extremely thin, they present a minimal barrier to gaseous exchange between the atmosphere and the blood in the capillaries. Adjacent alveoli are frequently in close contact; the intervening connective tissue forms the central part of the interalveolar septum. Alveolar macrophages are present within the alveolar lumen and migrate over the epithelial surface.

Interalveolar septum

The alveolar lining epithelium varies in thickness, but extensive areas of it are as little as 0.05 μm thick (see Fig. 54.13). The epithelium lies on a basal lamina that is fused with the basal lamina that surrounds the adjacent capillaries in the thin portions of a septum. The total barrier to diffusion between air and blood in these thin portions may be as little as 0.2 μm . The thick portions of a septum contain connective tissue elements, including elastic and collagen type III fibres, resident and migratory cells (see Fig. 54.12).

Alveolar epithelial cells (pneumocytes)

The alveolar epithelium is a mosaic of type I and II pneumocytes. Type I pneumocytes are simple squamous epithelial cells and form over 90% of the alveolar area. Their basal laminae fuse with that of the adjacent

Regional distribution of the cell types in the respiratory tract epithelium
(in approximate order of frequency in the region)

- Ciliated columnar epithelial cells
Goblet cells
Basal cells
Seromucous gland cells
Neuroendocrine cells
Brush cells
- Ciliated cuboidal epithelial cells
Basal cells
Seromucous gland cells (upper part of region only)
Brush cells
- Ciliated and non-ciliated cuboidal epithelial cells
Clara cells
Brush cells
- Ciliated and non-ciliated cuboidal epithelial cells
Clara cells
Type I pneumocytes
- Type I pneumocytes
Type II pneumocytes

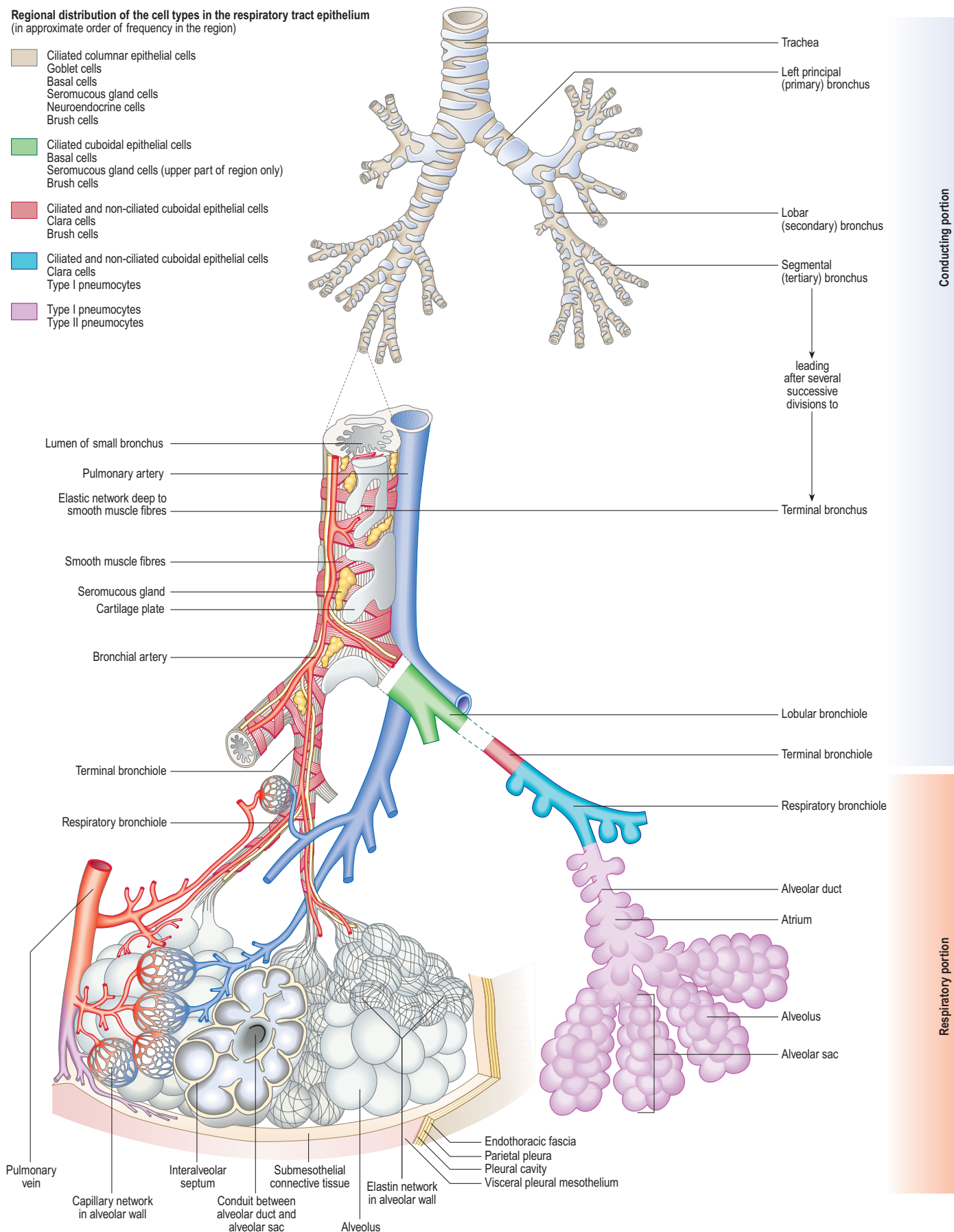


Fig. 54.11 The respiratory tree and its arterial blood supply, venous and lymphatic drainage, and nerve supply. The distribution of the different epithelial cells is shown. Blue is used for vessels that contain deoxygenated blood, and red for vessels that contain oxygenated blood.

capillary endothelium to form the thin portions of interalveolar septa. Their thin cytoplasm ($0.05\text{--}0.2\ \mu\text{m}$) facilitates gaseous diffusion between the lumen of the alveolus and its capillaries. The edges of adjacent cells overlap; they are joined by tight junctions, creating a strict diffusion barrier between the alveolar surface and underlying tissues.

Together with a similar endothelial barrier, this arrangement limits the movement of fluid from blood and interstitial spaces into the alveolar lumen (the blood–air barrier). If damaged, type I cells, which do not divide, are replaced by type II cells, which proliferate and differentiate into type I pneumocytes.

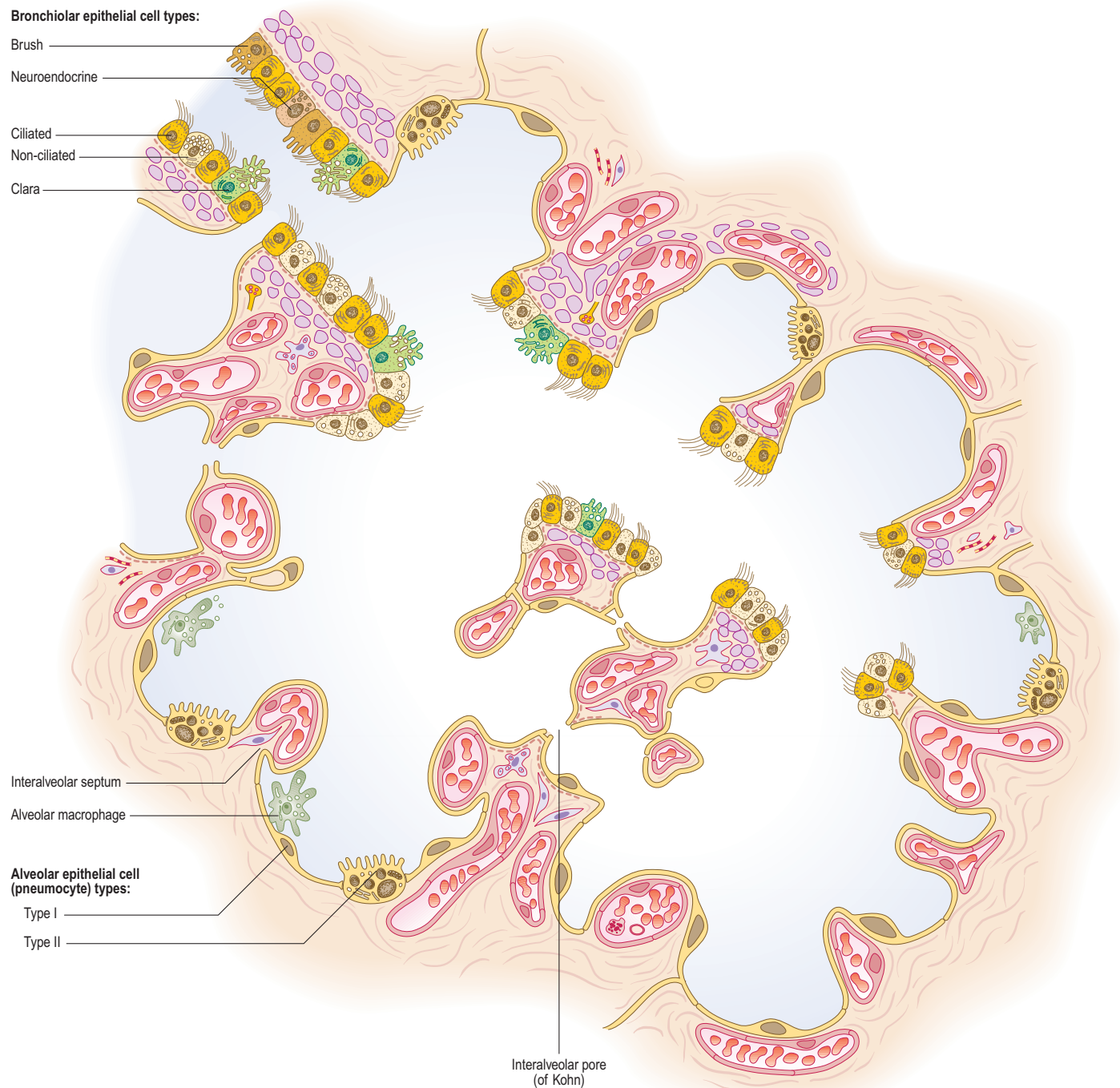


Fig. 54.12 The respiratory portion of the lung and its cell types.

The smaller type II cells are often more numerous than type I cells, but they contribute less than 10% of the alveolar surface area. They are rounded and protrude from the alveolar surface, particularly at the angles between alveolar profiles. In the human lung, they are often associated with interalveolar pores of Kohn. Their cytoplasm contains numerous characteristic secretory lamellar bodies consisting of concentric whorls of phospholipid-rich membrane, the precursors of alveolar surfactant, which they can recycle (see Fig. 54.13C).

Interalveolar pores of Kohn

Interalveolar pores of Kohn are small pores lined by epithelium (usually type II alveolar cells), which cross interalveolar septa to link adjacent alveolar airspaces. Humans have up to seven pores per alveolus, ranging in size from 2 to 13 μm . These small passages may sustain the flow of air in the event of blockage of one of the alveolar ducts and also provide routes of migration for alveolar macrophages, bacteria and viruses.

Alveolar macrophages

Alveolar macrophages are derived from circulating monocyte precursors (see Fig. 4.12). They originate in haemopoietic tissue in the bone marrow, migrate into the alveolar lumen from adjacent blood vessels and connective tissue, and wander about on the epithelial surfaces. They

clear the respiratory spaces of inhaled particles that are small enough to reach the alveoli (hence their alternative name of 'dust cells'; see Fig. 4.13). Most of them migrate with their phagocytosed load to the bronchioles, where they are swept into the mucociliary rejection current and removed from the lung. Others migrate through the epithelium of the alveoli into the lymphatics that drain the connective tissue of the lung, and so pass into lymphoid tissue around the pulmonary lobules. Under normal conditions, alveolar macrophages have a granular cytoplasm because they contain phagocytosed particles; in smokers, the particles have a characteristic appearance and are called 'tar bodies'. When actively phagocytic, macrophages release proteases: if antiproteases (e.g. α_1 -antitrypsin) that are normally present in the alveolar lining are deficient, macrophage activity may damage the lung. Alveolar macrophages are involved in the turnover of surfactant.

Alveolar macrophages can be recovered from sputum and are of diagnostic importance if they appear abnormal. For instance, whenever erythrocytes leak from pulmonary capillaries, the macrophages that engulf the extravasated cells become brick-red and are detectable in 'rusty' sputum. They are typical of congestive heart failure and often termed 'heart-failure cells'. Macrophages that migrate back into the connective tissue of the lung settle in patches that are visible beneath the visceral pleura, e.g. carbon-filled cells give the lungs a mottled

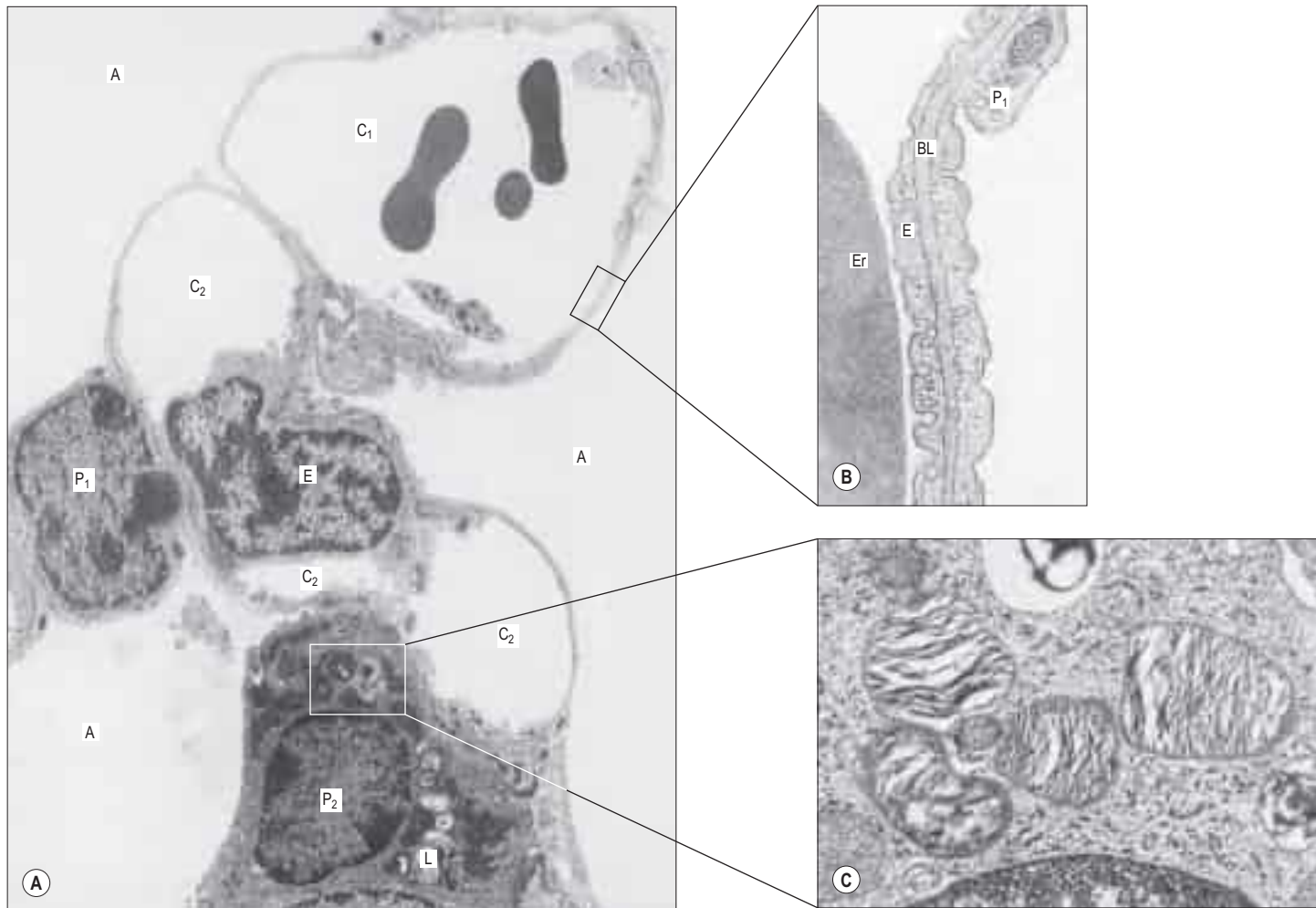


Fig. 54.13 **A**, An electron micrograph showing the septum between three adjacent alveoli (A). The septum contains two capillaries (C₁ and C₂) in section; the lower one is cut obliquely through the nucleus of an endothelial cell (E). Type I pneumocytes (P₁) line the alveolar airspaces except where a type II pneumocyte (P₂), with cytoplasmic lamellar bodies (L), intervenes. **B**, The thin portion of an interalveolar septum. Part of an erythrocyte (Er) is shown in the capillary lumen, lined by endothelium (E). The alveolar airspace (right) is lined by a type I pneumocyte (P₁). The two cells share a basal lamina (BL). **C**, Cytoplasmic lamellar bodies in a type II pneumocyte are composed mainly of phospholipids that contribute to the surfactant secreted by these pneumocytes and by Clara cells of the respiratory bronchioles. (With permission from Young B, Heath JW 2000 *Wheater's Functional Histology*. Edinburgh: Churchill Livingstone.)

appearance. However, if the inhaled particles are abrasive or chemically active, they may elude macrophage removal and damage the respiratory surface, producing fibrosis and a concomitant reduction in the respiratory area. This occurs in many industrial diseases, e.g. pneumoconiosis caused by coal dust, or asbestosis, where the long thin fibres of asbestos can cause considerable damage and may trigger fatal mesothelioma in the pleural lining.

Alveolar surfactant

The alveolar surface is normally covered by a film of pulmonary surfactant, which is a complex mixture mainly of phospholipids (particularly dipalmitoylphosphatidylcholine and phosphatidylglycerol), with some protein and neutral lipid. Surfactant is stored in lamellar bodies and secreted in the form of tubular myelin (unrelated to myelin of the nervous system) by type II pneumocytes. It is recycled by type II pneumocytes, or cleared (i.e. phagocytosed) by alveolar macrophages. Clara cells of the bronchiolar epithelium are believed to secrete surfactant of a different composition.

Surface tension at the alveolar surface is very high because the alveoli are minute and spherical; these features oppose expansion during inspiration, and tend to collapse the alveoli in expiration. The detergent-like properties of pulmonary surfactant greatly reduce the surface tension and make ventilation of the alveoli much more efficient. As the volume of an alveolus alternatively increases and decreases, so does its surface area. With a fixed amount of surfactant in the alveolar film, the relative concentration of surfactant increases on exhalation. The fortunate result is a decrease in collapsing tendency as the alveolar volume decreases, stabilizing the alveolus and overcoming the tendency of small gas spaces to obliterate.

TRACHEA AND BRONCHI

TRACHEA

The trachea is a 10–11 cm long tube formed of cartilage and fibromuscular membrane, and lined internally by mucosa. It descends from the larynx at the level of the sixth cervical vertebra and divides into right and left principal bronchi typically inferior to the sternal plane, level with the upper half of the sixth thoracic vertebra (see Fig. 51.6). It lies approximately in the sagittal plane but its point of bifurcation is usually a little to the right. The trachea is mobile and can rapidly alter in length, such that the bifurcation reaches the level of the sixth thoracic vertebra during deep inspiration. In children with full neck extension, the length of the trachea increases by 0.95 ± 0.43 cm; this elongation is insufficient to prevent displacement of the tip of an endotracheal tube towards the vocal cords in intubated children (Jin-Hee et al 2005).

The anterolateral portion of the trachea consists of 16–20 superimposed incomplete rings of hyaline cartilage and intervening fibroelastic tissue (the latter allows tracheal inspiratory elongation). The posterior aspect is a flat, fibromuscular wall. The trachea is funnel-shaped in neonates, when the upper end is wider than the lower end, and gradually becomes cylindrical with increasing age. The external transverse diameter is typically 2 cm in adult males and 1.5 cm in adult females. In the first postnatal year, the tracheal diameter does not exceed 4 mm. Mean transverse diameter is greater than anteroposterior diameter up to the age of 6 years and, after that, the two diameters are nearly equal (Griscom and Wohl 1986); during later childhood, tracheal diameter in millimetres is approximately equal to age in years. The lumen is about 12 mm in diameter in live adults but increases as a result of relaxation of the smooth muscle post mortem. The transverse shape of

the lumen is variable, especially in the later decades of life, and it may be round, lunate or flattened. At bronchoscopy, the posterior wall of the trachea bulges into the lumen and this is exaggerated in expiration and coughing. The distal end of the trachea is visible as a concave spur. A tracheal bronchus may occasionally arise, mainly from the right aspect of the trachea, as either a supernumerary or a displaced superior lobe bronchus (porcine bronchus). A tracheal bronchus was noted in 2.4% of children undergoing bronchoscopy for respiratory symptoms (Wong et al 1999). The variable position of the trachea in children makes it unreliable as an indicator of the presence of a right-sided aortic arch on a chest radiograph. In children with a right-sided aortic arch, the trachea is on the right in 47% of cases, in the midline in 33% and on the left in 7% (Strife et al 1989). Tracheal agenesis is a rare and usually lethal congenital abnormality.

Relations of the cervical part of the trachea

Anterior relations

The cervical trachea is covered anteriorly by skin and by the superficial and deep cervical fasciae. It is crossed by the jugular arch and overlapped by sternohyoid and sternothyroid. The second to fourth tracheal cartilages are crossed by the isthmus of the thyroid gland, above which an anastomosis connects the two superior thyroid arteries; anteroinferiorly are the pretracheal fascia, inferior thyroid veins, thymic remnants and the thyroid ima artery (when present). In children, the brachiocephalic trunk crosses obliquely in front of the trachea at, or a little above, the upper border of the manubrium; the left brachiocephalic vein may also rise a little above this level. An enlarged thyroid gland may cover the cervical trachea and reach the superior mediastinum, usually anterior to the left brachiocephalic vein.

Obstruction of the upper airways, craniofacial trauma and neck cancers (especially laryngeal), in either the acute or the chronic setting, require a tracheostomy: the creation of a hole (stoma) in the anterior aspect of the trachea that can serve independently or as a site for a dedicated tube allowing the spontaneous or mechanical ventilation of the patient. This manoeuvre usually involves the second and third cartilages; in particular cases, the lumen of the larynx can be approached (laryngotomy or cricothyroidotomy). Tracheostomy may be performed surgically or percutaneously (Ciaglia et al 1985, Griggs et al 1990).

Posterior relations

The oesophagus is a posterior relation of the cervical trachea, separating it from the vertebral column and the prevertebral fascia.

Lateral relations

The paired lobes of the thyroid gland, which descend to the fifth or sixth tracheal cartilage, and the common carotid and inferior thyroid arteries are lateral relations. The recurrent laryngeal nerves ascend on each side, in or near the grooves between the sides of the trachea and the oesophagus.

Relations of the thoracic part of the trachea

Anterior relations

As it descends through the superior mediastinum, the thoracic trachea lies behind the manubrium sterni, the origins of sternohyoid and sternothyroid, the thymic remnants and the inferior thyroid veins. The brachiocephalic and left common carotid arteries come to lie on the right and left of the trachea, respectively, as they diverge upwards into the neck. At a lower level, the aortic arch, brachiocephalic trunk, left common carotid artery, left brachiocephalic vein, deep cardiac plexus and lymph nodes are all anterior to the trachea.

Posterior relations

The oesophagus is posterior to the trachea and separates it from the vertebral column.

Lateral relations

Laterally and on the right are the right lung and pleura, right vagus nerve, right brachiocephalic vein, superior vena cava and azygos vein. On the left are the arch of the aorta, left common carotid and subclavian arteries. The left recurrent laryngeal nerve is at first situated between the trachea and aortic arch, and then lies within or just anterior to the tracheo-oesophageal groove.

The trachea and oesophagus can be compressed (or even eroded) by vascular rings resulting from abnormal formation of the aorta and its

branches, such as: double aortic arch, right aortic arch, aberrant right subclavian artery, double ductus arteriosus or persistence of the fifth aortic arch.

RIGHT PRINCIPAL BRONCHUS



The right principal bronchus is approximately 2.5 cm long, and is wider, shorter and more vertical than the left (see Fig. 54.7; Fig. 54.14A). These differences explain why inhaled foreign bodies enter the right principal bronchus more often than the left (more common in children who may present with breathlessness, unilateral wheeze or recurrent aspirations; a chest radiograph may show air trapping in the affected lobe). The right principal bronchus gives rise to its first branch, the superior lobar bronchus, then enters the right lung opposite the fifth thoracic vertebra. The azygos vein arches over it, and the right pulmonary artery lies at first inferior, then anterior to it (the eparterial bronchus). After giving off the superior lobar bronchus, which arises posterosuperior to the right pulmonary artery, the right principal bronchus crosses the posterior aspect of the artery, enters the pulmonary hilum posteroinferiorly, and divides into middle and inferior lobar bronchi.

Right superior lobar bronchus

The right superior lobar bronchus arises from the lateral aspect of the parent bronchus and runs superolaterally to enter the hilum; 1 cm from its origin it divides into three segmental bronchi.

Segmental anatomy

The apical segmental bronchus continues superolaterally towards the apex of the lung, which it supplies, and divides near its origin into apical and anterior branches. The posterior segmental bronchus serves the posteroinferior part of the superior lobe, passes posterolaterally and slightly superiorly, and soon divides into a lateral and a posterior branch. The anterior segmental bronchus runs anteroinferiorly to supply the rest of the superior lobe, and divides near its origin into a lateral and an anterior branch of equal size (see Fig. 54.7).

Bronchus intermedius

The bronchus intermedius is the continuation of the right principal bronchus distal to the right upper lobe bifurcation, coursing in the same direction as the former but with a slight rightward deviation. It is slightly flattened, 10 mm wide and 2.5 cm long. It begins at the base of the right upper lobe bronchus and gives rise to the common middle lobe and lower lobe bronchi, and sometimes to an additional variant subsuperior segmental bronchus (see below).

Right middle lobar bronchus

The right middle lobar bronchus usually starts 2 cm below the superior lobar bronchus and descends anterolaterally.

Segmental anatomy

The right middle lobar bronchus soon divides into a lateral and a medial segmental bronchus distributed to the homonymous segments of the middle lobe (see Fig. 54.7).

Right inferior lobar bronchus

The right inferior lobar bronchus is the continuation of the bronchus intermedius beyond the origin of the common middle lobar bronchus.

Segmental anatomy

At or a little below its origin from the principal bronchus, the right inferior lobar bronchus gives off a large superior (apical) segmental bronchus posteriorly. This runs posteriorly to the upper part of the inferior lobe, and then divides into medial, superior and lateral branches; the first two usually arise from a common stem. After giving off the superior segmental branch, the right inferior lobar bronchus descends posterolaterally. The medial basal segmental bronchus branches from its anteromedial aspect, and runs inferomedially to serve a small region below the hilum. The inferior lobar bronchus continues downwards and divides into an anterior basal segmental bronchus, which descends anteriorly, and a trunk that soon divides into a lateral basal segmental bronchus, which descends laterally, and a posterior

Normal variants in the bronchial anatomy are occasionally seen and consist of either displaced or supernumerary airways ([Ghaye et al 2001](#)). Abnormalities include a common origin of the right upper and middle lobe bronchi; an accessory cardiac bronchus; a right lower lobe bronchus that may arise from the left main stem bronchus; and an oesophageal bronchus. These anatomical variants are largely asymptomatic, occasionally causing haemoptysis, recurrent infection and bronchiectasis. Congenital bronchial atresia is associated with a bronchocele and emphysematous changes in the peripheral lung fields ([Wang et al 2012](#)). Left bronchial isomerism, characterized by two hyparterial bronchi and two bilobed lungs, and right bronchial isomerism, characterized by two eparterial bronchi and two trilobed lungs, may be features of the heterotaxy syndrome, which may include congenital cardiac, liver, stomach, intestinal and splenic abnormalities.

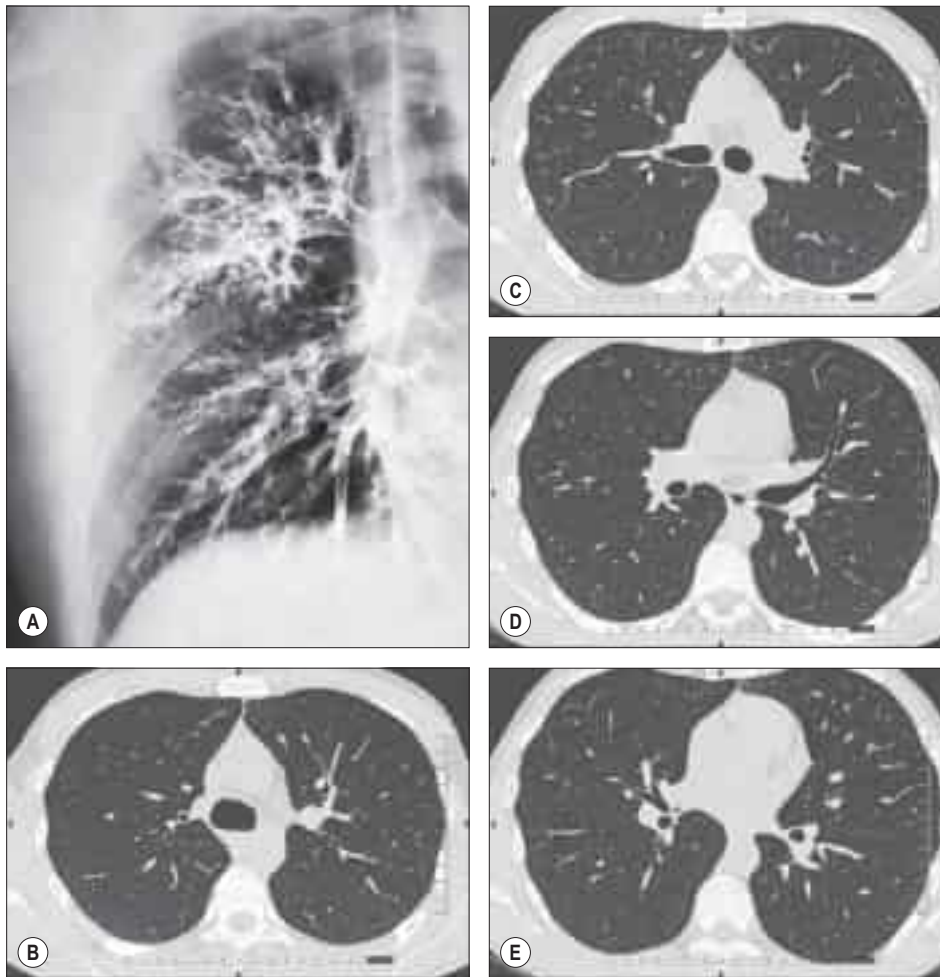


Fig. 54.14 **A**, A bronchogram showing the branching pattern of the trachea and bronchi of the right lung, in a slightly oblique, anteroposterior view. In this procedure, a radiopaque contrast medium has been introduced into the respiratory tract to coat the walls of the respiratory passages. For identification of the major branches, compare with **Figures 54.5B–E** and **54.6. B–E**, CT images demonstrating branching of lobar and segmental bronchi. These axial images correspond to level T4 (carina) to T6 (bifurcation right middle lobe bronchus), respectively.

basal segmental bronchus, which descends posteriorly. In more than half of all right lungs, a subsuperior (subapical) segmental bronchus arises posteriorly from the bronchus intermedius or right inferior lobar bronchus 1–3 cm below the superior segmental bronchus, and is distributed to the region of lung between the superior and posterior basal segments (see [Fig. 54.7](#)).

LEFT PRINCIPAL BRONCHUS

The left principal bronchus is narrower, less vertical and twice as long (5 cm) as the right principal bronchus. Passing to the left, inferior to the aortic arch, it crosses anterior to the oesophagus, thoracic duct and descending aorta; the left pulmonary artery is at first anterior and then superior to it (the hyparterial bronchus). It enters the hilum of the left lung at the level of the sixth thoracic vertebra and divides into superior and inferior lobar bronchi.

Left superior lobar bronchus

The left superior lobar bronchus arises from the anterolateral aspect of its parent stem, curves laterally, and soon divides into two bronchi that correspond to the branches of the right principal bronchus as it supplies the right superior and middle lobes, except that, on the left side, both are distributed to the left superior lobe because there is no separate middle lobe.

Segmental anatomy

The superior division of the left superior lobar bronchus ascends 1 cm, gives off an anterior segmental bronchus, continues a further 1 cm as the apicoposterior segmental bronchus, and then divides into apical and posterior branches. The apical, posterior and anterior segmental bronchi are largely distributed as they are in the right superior lobe. The inferior division descends anterolaterally to the anteroinferior part of the left superior lobe (the lingula) and forms the common lingular bronchus, which divides into superior and inferior lingular segmental bronchi. This is different from the pattern in the right middle lobe, where the corresponding distribution is lateral and medial (see [Fig. 54.7](#)).

Left inferior lobar bronchus

The left inferior bronchus descends posterolaterally and divides to supply territories of the lung that are distributed in essentially the same manner as they are in the right lung.

Segmental anatomy

The superior (apical) segmental bronchus arises from the inferior lobar bronchus posteriorly, 1 cm from its origin. After a further 1–2 cm, the inferior lobar bronchus divides into anteromedial and posterolateral stems. The latter divides into lateral and posterior basal segmental bronchi. The anterior basal segmental bronchus is an occasional independent branch of the inferior lobar bronchus, and a subsuperior (subapical) segmental bronchus occasionally arises posteriorly from the left inferior lobar bronchus (see [Fig. 54.7](#)).

ASPIRATION PNEUMONIA

Penetration of foreign substances into the bronchial tree, usually oral or gastric contents, produces aspiration pneumonia. Chemical (gastric acid) injury or bacterial infection complicates the clinical picture. The location depends on the position of the patient, since it is gravity-dependent; right middle and inferior lobes are the most common sites because of the larger and more vertical corresponding bronchi. Aspiration while standing leads to bilateral inferior lobar infiltrates. Patients lying down will aspirate in the right superior lobe (e.g. alcoholics).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Trachea

The trachea is supplied mainly by branches of the inferior thyroid arteries, which anastomose with ascending branches of the bronchial arteries. Veins draining the trachea end in the inferior thyroid venous plexus. The lymph vessels pass to the pretracheal and paratracheal lymph nodes.

Bronchi

Bronchial arteries

The bronchial arteries supply oxygenated blood and nutrition to the bronchi and connective tissue of the lungs. Each bronchial artery also has a branch that supplies the oesophagus. They are derived from the descending thoracic aorta either directly or indirectly, travel and branch with the bronchi, and terminate around the level of the respiratory bronchioles. They anastomose with the branches of the pulmonary arteries, and together supply the visceral pleura of the lung. Much of the blood supplied by the bronchial arteries is returned via the pulmonary veins rather than the bronchial veins (see below).

The single right bronchial artery has a variable origin, usually arising from the thoracic aorta as a common trunk with the right third posterior intercostal artery, sometimes from the superior left bronchial artery or any number of the right intercostal arteries, mostly the third right posterior. There are normally two left bronchial arteries (superior and inferior) that branch separately from the thoracic aorta.

The bronchial arteries accompany the bronchial tree and supply bronchial glands, the walls of the bronchi and larger pulmonary vessels. The bronchial branches form a capillary plexus in the muscular tunic of the air passages that supports a second mucosal plexus, which communicates with branches of the pulmonary artery and drains into the pulmonary veins. Other arterial branches ramify in interlobular loose connective tissue and most end in either deep or superficial bronchial veins; some also ramify on the surface of the lung, forming subpleural capillary plexuses. The bronchopulmonary arterial anastomoses in the walls of the smaller bronchi and in the visceral pleura may be more numerous in the neonate and are subsequently obliterated to a marked degree.

In addition to the three main bronchial arteries, smaller bronchial branches arise from the descending thoracic aorta; some may course through the pulmonary ligament and bleed during inferior lobectomy. The bronchial arteries are typically enlarged and tortuous in chronic pulmonary thromboembolic hypertension. They are not usually reconnected following lung transplantation. Aneurysms of a bronchial artery may mimic aortic aneurysms.

Some degree of mixing of blood between bronchial and pulmonary circulations and through Thebesian veins (*venae cordis minimae*) draining into the left ventricle normally occurs. A small degree of shunt, i.e. perfusion without ventilation, appears as normal and is usually 2–4%. Cardiac malformations with right-to-left flow, pulmonary arteriovenous malformations, congenital diaphragmatic herniae or pulmonary embolism cause more important shunting, which is refractory to administration of pure oxygen.

Bronchial veins

Usually two on each side, the bronchial veins drain blood from larger bronchi and hilar structures. They form two distinct systems. Deep bronchial veins commence as intrapulmonary bronchiolar plexuses that communicate freely with the pulmonary veins and eventually join a single trunk that ends in a main pulmonary vein or in the left atrium. Superficial bronchial veins drain extrapulmonary bronchi, visceral pleura and the hilar lymph nodes. They also communicate with the pulmonary veins and end in the azygos vein on the right, and in the left superior intercostal or the accessory hemiazygos vein on the left. Bronchial veins do not receive all the blood conveyed by bronchial arteries; some enters the pulmonary veins. The main bronchial arteries and veins run on the dorsal aspect of the extrapulmonary bronchi.

Lymphatic drainage

The deep lymphatic plexus reaches the hilum by travelling along the pulmonary vessels and bronchi. In larger bronchi, the deep plexus has submucosal and peribronchial parts, but in smaller bronchi there is only a single plexus that extends to the bronchioles. The walls of the alveoli have no lymphatic vessels.

INNERVATION

The anterior and posterior pulmonary plexuses innervate the trachea and bronchi. The two plexuses are interconnected. The trachea is innervated by branches of the vagi, recurrent laryngeal nerves and sympathetic trunks; the nerves enter the lung as networks that travel along branches of the bronchi and pulmonary and bronchial vessels as far as

the visceral pleura. Efferent vagal preganglionic axons synapse on small ganglia within the walls of the tracheobronchial tree, likely acting as sites of integration and/or modulation of the input from extrinsic nerves, or permitting local control of aspects of airway function by local reflex mechanisms.

MICROSTRUCTURE OF THE CONDUCTING AIRWAYS

The conducting airways are lined internally by a mucosa (see Fig. 2.16), and the epithelium lies on a thin connective tissue lamina propria. External to this is a submucosa, also composed of connective tissue, in which are embedded airway smooth muscle, glands, cartilage plates (depending on the level in the respiratory tree), vessels, lymphoid tissue and nerves. Cartilage is present from the trachea to the smallest bronchi, but is absent (by definition) from bronchioles.

Cartilaginous support

The trachea and extrapulmonary bronchi contain a framework of incomplete rings of hyaline cartilage, which are united by fibrous tissue and smooth muscle (see Fig. 54.16). Intrapulmonary bronchi contain discontinuous plates or islands of hyaline cartilage in their walls.

Tracheal cartilages

Each of the 16–20 tracheal cartilages measures about 4 mm in height and 1 mm in thickness, appearing thicker in the middle part; their external surface is flat while the internal aspect is slightly convex. The cartilaginous ‘rings’ encompass the anterior two-thirds of the trachea, the ring being completed in the posterior third by fibroelastic and muscular tissue. Cartilage loses elasticity and calcifies in older age, and adjacent cartilages may variously fuse. The first and the last two cartilages are more conspicuous, the first being the thickest (see Fig. 54.7). The last tracheal ring merges into the incomplete rings at the origin of each principal bronchus; the bifurcation is marked by a cartilaginous spur, the carina (see Fig. 54.11).

Bronchial cartilages

Cartilaginous plates mainly surround the dorsal aspect of the lobar bronchi. They become more irregular and form progressively less of the bronchial walls in intrapulmonary bronchi, and disappear in the bronchioles.

PULMONARY DEFENSIVE MECHANISMS

The respiratory tract presents a huge surface area that is vulnerable to desiccation, freezing, microbial invasion and the mechanical and chemical effects of inhaled particles. Inhaled air is humidified chiefly in the upper respiratory tract, where it passes, with some turbulence, over the nasal and buccopharyngeal mucosae. The secretions of the various glands of the bronchial tree also help to prevent desiccation. Goblet cells secrete sulphated acid mucosubstances; cells in mucous glands beneath the epithelial surface contain mainly carboxylated mucosubstances, particularly those associated with sialic acid, although sulphated groups also occur; cells of serous glands contain neutral mucosubstances. Goblet cells respond mainly to local irritation, while tubular glands, both mucous and serous, are mainly under neural and hormonal control. Excessive or altered secretions may obstruct the flow of air. Antibacterial and antiviral substances, e.g. lysozyme, IgA antibodies and possibly interferon, also appear in the secreted fluid.

Inhaled particles may be removed via the mucociliary rejection current. Cilia sweep the fluid overlying the surfaces of bronchioles, bronchi and trachea upwards at 1 cm/min, and much of the inhaled matter trapped in the viscous fluid may be removed in this way. Particles small enough to reach the alveoli may be removed by alveolar phagocytes. Alveolar epithelium has limited powers of regeneration but normally is continually replaced. The lifespan of alveolar squamous cells is approximately 3 weeks, and that of alveolar phagocytes is 4 days. Inhaled particles may also be cleared by the cough reflex.

Numerous lymphoid nodules occur in the bronchial lining, providing foci for the production of lymphocytes and giving local immunological protection against infection both by cell-mediated (T-cell) activities, and by the production of immunoglobulins (mainly IgA) from B cells, which are passed on to gland cells for secretion to the epithelial surface.

The predominant contractile innervation of airway smooth muscle is parasympathetic and cholinergic in nature, while the primary relaxant innervation of the airways is non-cholinergic (parasympathetic nerves containing nitric oxide synthase and vasoactive intestinal peptide). These parasympathetic nerves are anatomically and physiologically distinct from one another and differentially regulated by reflexes. Sympathetic-adrenergic nerves play little, if any, role in directly regulating smooth muscle tone in the human airways. Activation of airway afferent nerves (rapidly adapting receptors, C fibres) can evoke increases in airway smooth muscle parasympathetic nerve activity, or decreases in parasympathetic nerve activity (via activation of slowly adapting receptors). Extrapulmonary afferents may modulate neurally mediated regulation of airway smooth muscle tone (Canning 2001a).

Stimulation of cholinergic nerves causes not only bronchoconstriction but also mucus secretion and bronchial vasodilation. Sympathetic nerves may control tracheobronchial blood vessels.

Epithelium

The epithelia of the trachea, bronchi and bronchioles are, in general, similar to each other, with graded variations in the numbers of different cell types. The extrapulmonary and larger intrapulmonary passages are lined with respiratory epithelium (see Fig. 2.2D), which is pseudostratified and predominantly ciliated, and contains interspersed mucus-secreting goblet cells. There are fewer cilia in terminal and respiratory bronchioles, and the cells are reduced in height to low columnar or cuboidal. The epithelium of smaller bronchi and bronchioles is folded into conspicuous longitudinal ridges, which allow for changes in luminal diameter (see Fig. 54.11). The epithelium in the respiratory bronchioles progressively reduces in height towards the alveoli, and is eventually composed of cuboidal, non-ciliated cells. Respiratory bronchioles have lateral pouches in their walls, which are lined with squamous cells, so providing an accessory respiratory surface.

Six distinct types of cell have been described in the conducting airways, namely: ciliated columnar, goblet, Clara, basal, brush and neuroendocrine (see Fig. 54.11). Lymphocytes and mast cells migrate into the epithelium from the underlying connective tissue.

Ciliated columnar cells

Ciliated columnar cells are the driving force of the mucociliary rejection current (escalator) in the bronchial tree. They vary from low to tall columnar, and each has up to 300 cilia (Ch. 1, see Fig. 1.17) projecting from the apical surface. The cilia extend into a watery fluid secreted by serous cells of the submucosal glands, but their tips are in contact with a more superficial layer of thicker mucus secreted by surface goblet cells and mucous cells in the submucosal glands. The rate of ciliary beating is usually 12–16 per second; mechanical stimulation of the epithelial surface and inflammatory mediators increase the rate. In addition to tight junctions, which seal the apical intercellular space from the airway lumen, the ciliated cells are coupled by gap junctions, which allow a change in rate of beating to spread from stimulated cells to their neighbours (probably via calcium signalling) so that their metachronal coordination remains intact. Primary ciliary dyskinesia is a ciliary abnormality inherited as an autosomal recessive disorder and may be a feature of Kartagener's syndrome, which is a combination of bronchiectasis, sinusitis and situs inversus.

Goblet cells

Goblet cells are present from the trachea (6000–7000 per mm²) distal to the smaller bronchi, but are normally absent from bronchioles. They contain an apical region full of large secretory vacuoles filled with mucinogen. When the epithelium is irritated, e.g. by tobacco smoke, there is an overall increase in the number of goblet cells, and they also extend into the bronchioles.

Clara cells

Clara cells are cuboidal, non-ciliated cells with apices that bulge into the lumen. They contain numerous electron-dense secretory granules and many lysosomes. Clara cells produce surfactant lipoprotein and thus share functional similarities with the type II alveolar cell of pulmonary alveoli, although their secretory granules differ in structure and composition. They may also regulate ion transport.

Basal cells

Basal cells are small, rounded cells that occur in parts of the airway lined by pseudostratified respiratory epithelium; they are mitotic stem cells for other epithelial cell types. Basal cells contact the basal lamina and are most frequent in the larger conducting passages.

Brush cells

Brush cells are slender, non-ciliated cells with characteristically long, stiff apical microvilli, from which they derive their name. Although infrequent, they are present throughout all parts of the conducting airway passages, including the respiratory epithelium of the nasopharynx. They are in contact with afferent nerve fibres basally and are presumed to have a sensory receptor function.

Neuroendocrine cells

Neuroendocrine cells, also termed dense-core or small granule (Kulchitsky) cells, are found mainly in the basal part of the epithelium. They have a rounded shape and the cytoplasm basal to their nuclei contains numerous small, dense-cored vesicles, approximately 150 nm in diameter. They form part of the dispersed neuroendocrine system of amine precursor uptake and decarboxylation (APUD) cells. Neuroendocrine cells are most numerous in fetal lungs and their number decreases dramatically after birth. There appears to be little further change in their frequency during adult life, although they may proliferate in certain pulmonary diseases.

Lymphocytes

T lymphocytes, derived from mucosa-associated lymphoid tissue (MALT) in the walls of the passages, occur within the epithelium throughout the conducting airway tissues, particularly in the extrapulmonary portion. They are concerned with the immune surveillance of the epithelium. Clusters of lymphocytes sometimes lie beneath non-ciliated epithelial cells of the microfold (M-cell) type.

Mast cells

Mast cells are present within the basal regions of the epithelium. They resemble connective tissue mast cells, and their cytoplasmic histamine-containing granules are released in response to irritants, including inhaled allergens.

Submucosal glands

Tubuloacinar, seromucous glands are present in the submucosa of the trachea and bronchi and, to a lesser extent, in the larger bronchioles. They contain separate mucous and serous cells and are an important source of the mucus at the surface of the ciliated respiratory epithelium. Their secretions include mucins; bacteriostatic substances such as lysozyme and lactoferrin; secretory antibodies (immunoglobulin A (IgA)) produced by plasma cells in the submucosal connective tissue; and protease inhibitors (e.g. α_1 -antitrypsin), important for neutralizing leukocyte-derived proteases in the respiratory tract, particularly elastase. Deficiency of α_1 -antitrypsin causes chronic obstructive pulmonary disease, by inducing panacinar emphysema and bronchiectasis. The secretory acini and tubules are surrounded by myoepithelial cells, which are innervated by autonomic fibres (see above).

Connective tissue and muscle

Broad, longitudinal bands of elastin within the submucosa (Fig. 54.15) follow the course of the respiratory tree and connect with the elastin networks of the interalveolar septa (see Fig. 54.11). This elastic framework is a vital mechanical element of the lung and is responsible for elastic recoil during expiration.

In the trachea and extrapulmonary bronchi, the smooth muscle is mainly confined to the posterior, non-cartilaginous part of the tracheal tube (Fig. 54.16). Along the entire intrapulmonary bronchial tree, smooth muscle forms two opposed helical tracts, which become thinner and finally disappear at the level of the alveoli. The tone of these muscle fibres is under nervous and hormonal control; groups of muscle cells are coupled by gap junctions to spread excitation within fascicles.

Muscle cell contraction narrows the airway, while relaxation permits bronchodilation. Some tone normally exists in the muscular bands, which relax slightly during inspiration and contract during expiration, thereby assisting the tidal flow of air. Abnormal contraction may be caused by circulating smooth muscle stimulants or by local release of excitants such as 5-HT (5-hydroxytryptamine, serotonin), histamine and leukotrienes, which produces bronchospasm. Numerous mast cells are present in the connective tissue of the respiratory tree, especially towards the bronchioles.

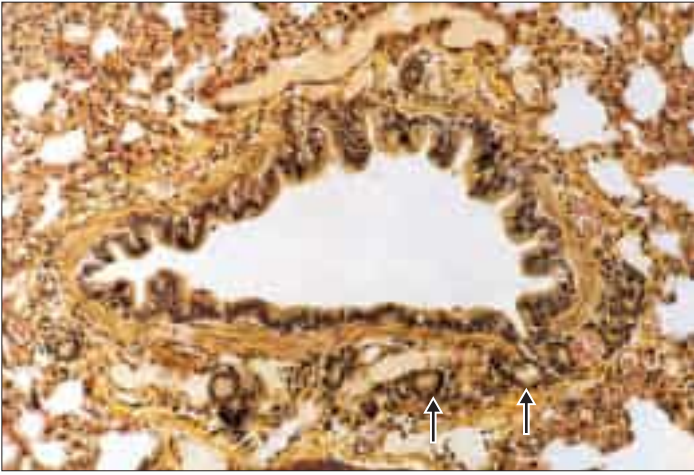


Fig. 54.15 A large bronchiole stained for elastin (black), showing smooth muscle in the bronchiolar wall and a few small seromucous glands (arrows).



Fig. 54.16 A micrograph of the trachea of a child, sectioned transversely. An incomplete cartilage ring (C) is joined posteriorly by muscular tissue (M). Seromucous submucosal glands (S) are concentrated in this region and between adjacent cartilages. Note the thyroid tissue (bottom). (With permission from Stevens A, Lowe JS 1996 Human Histology, 2nd ed. London: Mosby.)

Anatomy of coughing

Coughing is an important defensive reflex. It enhances clearance of secretions and particulates from the airways and protects from aspiration of foreign materials (occurring as a consequence of aspiration or inhalation of particulate matter, pathogens, accumulated secretions, postnasal drip, inflammation and mediators associated with inflammation).

Under normal conditions, cough serves an important protective role in the airways and lungs, but in some conditions it may become excessive, non-productive and potentially harmful to the airway mucosa.

The mechanical events of a cough can be divided into three phases. An initial deep inhalation generates the volume necessary for an effective cough and is followed by a compression phase, when forceful contraction of the muscles of the chest wall, diaphragm and abdominal wall against a closed glottis results in a rapid rise in intrathoracic pressure. Subsequent glottal opening causes a rapid peak expiratory airflow that dislodges and expels mucus from the airways; forceful airway compression produces the coughing sound.

BRONCHOSCOPY

Bronchoscopy allows the direct visualization of the vocal cords, trachea and major airways as far as the first division of the subsegmental airway. Occasionally, bronchoscopy can also provide some information about structures adjacent to the airways, e.g. significant subcarinal lymphadenopathy may cause splaying or widening of the carina. Bronchoscopy enables the acquisition of samples (e.g. bronchial lavage, bronchial brushings, bronchial and transbronchial biopsies); provides information about staging in lung cancer; and facilitates therapeutic procedures (e.g. foreign body removal, ablation of tumours and insertion of airway stents). The information obtained by volume CT scanning can be reconstructed to provide three-dimensional images that can be used to create endobronchial views (virtual bronchoscopy, [Fig. 54.17](#)) or to demonstrate the pulmonary arteries.

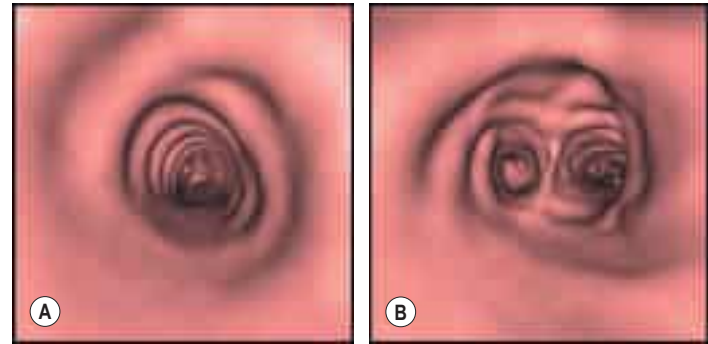


Fig. 54.17 A virtual CT bronchoscopic view obtained by planar reconstruction of spiral CT. **A**, Left principal bronchus. **B**, Carina. (Courtesy of GE Medical Systems.)



Bonus e-book images

Fig. 54.9 A lung window axial view showing an intralobar sequestration in the left lower lobe of the lung in a 15-month-old girl.

Fig. 54.10 The systemic arterial blood supply from the abdominal aorta to an intralobar sequestration in the left lower lobe of the lung in a 15-month-old girl.

Fig. 54.15 A large bronchiole stained for elastin, showing smooth muscle in the bronchiolar wall and a few small seromucous glands.

Fig. 54.16 A micrograph of the trachea of a child, sectioned transversely.

Coughing is initiated when mechanically and chemically sensitive vagal afferents innervating the airways from the upper respiratory tract to the terminal bronchioles and lung parenchyma are activated (Morice and Gepetti 2004). Data from experimental studies suggest that the receptors innervated by these afferents may be broadly divided into three groups: namely, slowly (SAR) and rapidly (RAR) adapting stretch receptors and bronchopulmonary unmyelinated C fibres. The vagal afferents terminate centrally in largely non-overlapping regions of the caudal half of the nucleus tractus solitarius. Second-order neurones from this nucleus terminate on respiratory-related regions of the pons, medulla and spinal cord. Efferent impulses travel via the vagus, phrenic and spinal motor nerves to the inspiratory and expiratory muscles, larynx, diaphragm and abdominal wall muscles. Airway receptors in the external auditory canals, eardrums, paranasal sinuses, pharynx, diaphragm, pleura, pericardium and stomach are probably mechanoreceptors, stimulated by triggers such as touch or displacement (Polverino et al 2012).

Rapidly adapting receptors (RAR) RARs are dynamic receptors that respond to changes in airway mechanical properties (e.g. diameter, length and interstitial pressures). Functional studies of RARs suggest that they terminate within or beneath the epithelium mainly of the intrapulmonary airway (Widdicombe 2003). They may be differentiated from other airway afferents by their rapid adaptation (in 1–2 seconds) to sustained lung inflation (Schelegle and Green 2001). RARs are sporadically active throughout the respiratory cycle, activated by the dynamic mechanical forces that accompany lung inflation and deflation, and become more active as the rate and volume of lung inflation increase. Their activation initiates reflex bronchospasm and mucus secretion through parasympathetic pathways.

Slowly adapting stretch receptors (SARs) SARs are highly sensitive to the mechanical forces that act on the lung during breathing. Their activity increases during inspiration and peaks just prior to the initiation of expiration. They are thus thought to be the afferent fibres involved in the Hering–Breuer reflex, which terminates inspiration and initiates expiration when the lungs are adequately inflated. SARs appear to terminate primarily in the intrapulmonary airways. SAR activation results in the central inhibition of respiration and the inhibition of the cholinergic drive to the airways, leading to decreased phrenic nerve activity and decreased airway smooth muscle tone (Canning et al 2001b); most of these SARs are associated with alveoli or bronchioles.

C fibres The majority of afferent nerves innervating the airways and lungs are unmyelinated C fibres, similar to the unmyelinated somatic sensory nerves that innervate the skin, skeletal muscle, joints and bones and that respond to noxious chemical and mechanical stimuli. In addition to their slow conduction velocity (<2 m/s), airway vagal afferent C fibres are distinguished from RARs and SARs by their relative insensitivity to mechanical stimulation and lung inflation. They are generally quiescent throughout the respiratory cycle (Lee and Pisarri 2001).

The specific pattern of the cough depends on the site and type of stimulation. Mechanical laryngeal stimulation results in immediate

expiratory stimulation (sometimes termed the expiratory reflex), protecting the airway from aspiration; stimulation distal to the larynx causes a more prominent inspiratory phase to generate the airflow necessary to remove the stimulus.

During vigorous coughing, intrathoracic pressures may reach 300 mmHg (with an abrupt rise in pleural pressure from 6.5 to 13 kPa) and expiratory velocities approach 500 miles per hour (Ford et al 2007). While these pressures and velocities are responsible for the beneficial effects of cough on mucus clearance, they are also the cause of many of the complications of cough, including exhaustion, self-consciousness, insomnia, headache, dizziness, musculoskeletal pain, hoarseness, excessive perspiration, urinary incontinence and cough-induced multiple rib fractures.

Lesions that compress the upper airway, including arteriovenous malformations and retrotracheal masses, may present with chronic cough. Cough can also be a symptom of tracheobronchomalacia, which results from the loss of rigid support of the large airways and inspiratory collapse, usually seen in conjunction with obstructive lung disease in patients with a history of cigarette smoking.

Chronic dry cough may be caused by irritation of the external auditory canal by impacted foreign bodies or cerumen (Jegoux et al 2002). The aetiology of the 'ear-cough' (otorespiratory reflex) is related to stimulation of the auricular branch of the vagus nerve (Arnold's nerve) (Feldman and Woodworth 1993). Holmes–Adie syndrome is a rare cause of coughing when autonomic dysfunction affects the vagus nerve. Patients present with anisocoria, abnormal deep tendon reflexes and patchy areas of hyperhidrosis or anhidrosis (Kimber et al 1998).

Differences among several sites from which cough stimuli can originate may result in variations in the sounds and patterns of coughing. Laryngeal stimulation produces a choking type of cough without a preceding inspiration. Inadequate mucociliary clearance mechanisms (as in bronchiectasis or cystic fibrosis) may produce a pattern of coughing with less violent acceleration of air and a sequence of interrupted expirations without any intervening inspiration.

Since cough is an important defensive reflex required to maintain the health of the lungs, people who do not cough effectively are at risk of atelectasis, recurrent pneumonia and chronic airways disease from aspiration and retention of secretions. Many disorders can impair the ability to cough effectively, which may result in persistent cough. The elderly, the newborn, lung transplant recipients and patients with paralysis or neuromuscular disorders have a poorly developed and/or compromised cough reflex, and are rendered highly susceptible to infective and aspiration pneumonia (Polverino et al 2012). Patients with paralysis or neuromuscular disorders (including rib fractures) and chest wall deformities may not generate the flows necessary for effective clearance of secretions due to defective 'pump' mechanisms. Individuals with reduced function of the abdominal wall musculature are particularly at risk of ineffective cough. Patients with tracheobronchomalacia ('floppy' airways) or with obstructive airways diseases often do not generate the high flow rates needed for effective clearance of secretions. Individuals with laryngeal disorders, including those with tracheostomies, may not achieve sufficient laryngeal closure to generate the increased intrathoracic pressures necessary for an effective cough.

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